# Protocol compiler (protocol-compiler)

Statements digest, version 0.24.0

Every definition, fact, assumption, result and requirement of the contract, with the contract's numbers; remarks, examples, technical steps and all proofs are left out. Each heading opens its statement in the contract.

# Mathematics

## 1 The protocol declaration

**Definition 1.1** (Protocol declaration)**.** A *protocol declaration* is a tuple

$$\mathfrak{D} = \Big(A,\ \tau_A;\ Q,\ \mathbb{S};\ \{M_x\}_{x\in\mathcal{X}};\ \{F_v\}_{v\in\mathcal{V}};\ \mathfrak{P};\ a_A,\ a_B;\ \phi;\ r,\ \mathsf{h};\ \mathfrak{L}\Big), \quad (1.1)$$

with components as follows. $A$ and $Q$ are finite-dimensional systems, $\mathcal{H}(Q) = \bigoplus_{s\in\mathbb{S}} \mathcal{H}(Q_s)$ with a finite index set $\mathbb{S}$ of sectors, sector projectors $\Pi_s$ and dimensions $d_s = \dim \mathcal{H}(Q_s)$, $d_Q = \sum_s d_s$, $d_A = \dim \mathcal{H}(A)$. The sector map $\mathfrak{s}$ is the ordered list $\big((s, d_s)\big)_{s\in\mathbb{S}}$ of block identifiers with their dimensions, together with its block kind, a label naming what the identifiers mean (Definition "Sector maps and block-list operators" (foundations/def:sector-map)). It is one object, carried with the declaration beside the slot $Q, \mathbb{S}$ of (1.1), and it is what fixes $\mathbb{S}$, its total order and $(d_s)_s$. The *marginal slot* holds either a state $\tau_A \in S_=(A)$, in trusted-source mode, or the value none, in untrusted-source mode. $\{M_x\}_{x\in\mathcal{X}}$ is a POVM on $A$, given as one matrix per $x$, since $A$ carries no sector decomposition. The words "state" and "POVM" in the two sentences above name the intended objects, and they are also exactly the conditions this contract decides: what a declaration is required to satisfy is D(i)–D(viii) of Definition 1.5, every operator predicate of which is decided *exactly* on the declared binary64 numbers, together with the summation condition of Definition 1.3, without which a computed sum would not be the sum it is read as.

*Operators on $Q$ are block lists.* An operator on $Q$ is given as one array $F^{(s)} \in \mathcal{L}(Q_s)$ per $s \in \mathbb{S}$, against the sector map $\mathfrak{s}$, and that list is the operator: it is a block list against $\mathfrak{s}$, the operator it denotes is $\bigoplus_s F^{(s)}$, and the declaration holds no array larger than a single sector block. Where $|\mathbb{S}| = 1$, say $\mathbb{S} = \{s_0\}$, that one block *is* the whole of $Q$ and is held and emitted as such, $d_{s_0} = d_Q$, as on the Band-A route (Theorem 8.2 (i)). Sector block-diagonality is therefore a property of the *type*: there is no off-sector entry an operator of this form could carry, and correspondingly nothing to verify. So $\{F_v\}_{v\in\mathcal{V}}$ is a family of such block lists over a finite alphabet, one list per $v$; $F_v^{(s)}$ denotes its $s$-block. A state $\omega$ on $AQ$ is likewise a list, one

array $\omega^{(s)} \in \mathcal{L}(AQ_s)$ per $s$, and $\omega = \bigoplus_s \omega^{(s)}$ in the same sense. $\mathfrak{P}$ is a *party partition*: two total maps $\pi_{\mathsf{A}} : \mathcal{X} \times \mathcal{V} \to \mathcal{O}_{\mathsf{A}}$ and $\pi_{\mathsf{B}} : \mathcal{X} \times \mathcal{V} \to \mathcal{O}_{\mathsf{B}}$ onto declared finite alphabets, each a *readout map*, together with a party label for $A$ and, for each $s$, a declared tensor factorization of $\mathcal{H}(Q_s)$ with a party label per factor. $a_A : \mathcal{O}_{\mathsf{A}} \to \mathcal{C}_A$ and $a_B : \mathcal{O}_{\mathsf{B}} \to \mathcal{C}_B$, each an *announcement map*, are written jointly

$$a(x, v) \coloneqq \big(a_A(\pi_{\mathsf{A}}(x, v)),\, a_B(\pi_{\mathsf{B}}(x, v))\big) \in \mathcal{K} \coloneqq \mathcal{C}_A \times \mathcal{C}_B, \tag{1.2}$$

$\phi : \mathcal{K} \to \widehat{\mathcal{C}}$ is the acceptance-statistic map, $r : \mathcal{X} \times \mathcal{V} \to \mathcal{Z}_\perp$ the raw-key map with $\mathcal{Z}_\perp = \mathcal{Z} \sqcup \{\perp\}$ and $\mathcal{Z} = \{0, \dots, d_Z - 1\}$, $\mathsf{h} \in \{\mathsf{A}, \mathsf{B}\}$ the declared key holder, and $\mathfrak{L} = \{(\hat{O}_k, \bowtie_k, \mathrm{rhs}_k)\}_{k=1}^{m}$, $m \geq 0$, a finite constraint list, each entry an operator $\hat{O}_k$, a relation symbol $\bowtie_k$ and a right-hand side $\mathrm{rhs}_k$ (which symbols a well-formed declaration admits is D(vi) of Definition 1.5), with every $\hat{O}_k$ a block list against the same $\mathfrak{s}$, Hermitian block by block.

*The right-hand side of an entry carries a declared tag, and the tag is part of the datum rather than a property measured of it.* In trusted-source mode an entry is declared either homogeneous, carrying a real scalar $c_k$, from which the right-hand side it denotes is *constructed* as $\Theta_k \coloneqq c_k \tau_A$, or general, carrying a Hermitian matrix $\Theta_k$ on $A$. In untrusted-source mode only the scalar form exists and the right-hand side is a real number $\Theta_k = c_k$. So an entry is written $(\hat{O}_k, \bowtie_k, c_k)$ exactly when it is declared homogeneous or is untrusted-mode, and $(\hat{O}_k, \bowtie_k, \Theta_k)$ when it is declared general (Definition 2.5). For these two forms the tag is the type of the right-hand-side datum, a real scalar or a Hermitian matrix, which is why Definition 2.5 calls them the untagged forms; the third form, party-local, exists only in the general format below and is declared by its tag (DX6). We write $\kappa = (c_A, c_B) \in \mathcal{K}$, a value of the joint announcement (1.2), for an announcement class and

$$\widehat{\mathcal{C}}_{\mathrm{key}} \coloneqq \big\{\hat{c} \in \widehat{\mathcal{C}} :\ \exists\, \kappa \in \phi^{-1}(\hat{c}) \text{ with } \varsigma(\kappa) = \mathrm{pass}\big\} \tag{1.3}$$

for the *key-live* acceptance values. Here $\varsigma$ is the sifting predicate of the specification's public part (Definition 3.1 (d)), read as $\varsigma(\kappa) = \mathrm{fail}$ on every class $\kappa \notin a(\mathcal{X} \times \mathcal{V})$, which no outcome pair realises; on the realised classes it is, under the hypotheses of Theorem 5.1, a witness of D(v) of Definition 1.5 (Theorem 5.1 (ii)). It is *not* a slot of (1.1), so (1.3) is the set of acceptance values to which some realised class passing sifting maps, fixed by the specification and not by the declaration; the key-live set of the consuming engine, Definition "The protocol declaration as the MEAT core reads it" (meat-core/def:declaration), takes the same convention on the classes no outcome pair realises.

*The general format.* Everything above is written for the format in which $A$ carries one sector, one system is measured, Alice's measurement is declared as the POVM $\{M_x\}$, the public discussion is two messages neither of

which reads the other, and no entry is declared party-local; we call it the *basic format*. It is the case $|\mathbb{T}| = 1$, $|\mathbb{F}| = 1$, POVM form, $L = 2$ of the following one, *entry for entry* on every component a derivation reads: at those values every such object defined in (DX1)–(DX6) is the object of the same name above, and nothing else is declared. On the party labels of the tensor factors of each sector block, which no derivation of this module reads and which the consuming engine's laboratory premise reads as naming the parties' laboratories, the general format is narrower than the basic one. In general a declaration is the tuple above with its slots read as follows.

(DX1) *(factorised sector map)* The sector map is an ordered list, each entry of which is a *factor*. Each factor carries a *system tag*, $A$ or $Q$, a party label, and a sector map of its own, an ordered list of block identifiers with their dimensions; all factors carry one block kind. At most one factor is tagged $A$, and where there is one it is the first of the list; at least one factor is tagged $Q$. The factor tagged $A$ fixes a finite ordered index set $\mathbb{T}$ and a decomposition $\mathcal{H}(A) = \bigoplus_{t\in\mathbb{T}} \mathcal{H}(A_t)$ into subspaces spanned by vectors of the declared basis of $A$, with projectors $\Pi_t^A$ and dimensions $d_{A_t}$, $d_A = \sum_t d_{A_t}$; where no factor is tagged $A$, the system $A$ is still present and is one Alice sector, $\mathbb{T} = \{t_0\}$ and $A_{t_0} = A$. A factor tagged $Q$ is a *measured factor*, and the measured factors form the ordered set $\mathbb{F}$; factor $f$ fixes a system $Q_f$ with $\mathcal{H}(Q_f) = \bigoplus_{s_f\in\mathbb{S}^f} \mathcal{H}(Q_{f,s_f})$, projectors $\Pi_{s_f}^f$ and dimensions $d_{f,s_f}$, and for $s = (s_f)_f$

$$Q = \bigotimes_{f\in\mathbb{F}} Q_f, \qquad \mathbb{S} = \prod_{f\in\mathbb{F}} \mathbb{S}^f, \qquad \mathcal{H}(Q_s) = \bigotimes_f \mathcal{H}(Q_{f,s_f}), \qquad \Pi_s = \bigotimes_f \Pi_{s_f}^f, \qquad d_s = \prod_f d_{f,s_f} \tag{1.4}$$

all products in the order of $\mathbb{F}$, and $\mathbb{S}$ ordered with the last factor fastest, a product identifier being the factors' identifiers joined in that order. Each pair $(t, s) \in \mathbb{T}\times\mathbb{S}$ is a *joint sector*; the joint sectors are ordered with $t$ slowest, and $\mathcal{H}(AQ) = \bigoplus_{(t,s)} \mathcal{H}(A_t)\otimes\mathcal{H}(Q_s)$, the $A$ index first inside a block. The party partition labels $A$ with the party of the factor tagged $A$, where there is one, and otherwise with the party label it declares for $A$, as in the basic format; and it labels every tensor factor of $\mathcal{H}(Q_{f,s_f})$ with the party of $f$.

(DX2) *(Alice's data and $\omega$ are block lists)* $\tau_A$, every operator of Alice's measurement slot (DX4) and the right-hand side $\Theta_k$ of a general entry are lists of one array on $\mathcal{H}(A_t)$ per $t \in \mathbb{T}$, and the list is the operator: $\tau_A = \bigoplus_t \tau_A^{(t)}$, $M_x = \bigoplus_t M_x^{(t)}$, $\Theta_k = \bigoplus_t \Theta_k^{(t)}$. A state $\omega$ on $AQ$ is a list of one array $\omega^{(t,s)} \in \mathcal{L}(A_tQ_s)$ per joint sector, $\omega = \bigoplus_{(t,s)} \omega^{(t,s)}$. Block diagonality along $\mathbb{T}$ is a property of the type: no off-sector entry can be written. By itself this typing licenses nothing; whether restricting $\omega$ to such lists is sound is not settled by this definition.

(DX3) *(factorised family)* The outcome alphabet is the product $\mathcal{V} = \prod_{f\in\mathbb{F}} \mathcal{V}^f$ of the factors' ordered alphabets, in the product order, the last factor fastest. What is declared is one family of block lists $\{F^f_{v_f}\}_{v_f\in\mathcal{V}^f}$ per measured factor, against that factor's own sector map, together with the rule

$$F_v \coloneqq \bigotimes_{f\in\mathbb{F}} F^f_{v_f}, \qquad F^{(s)}_v = \bigotimes_{f\in\mathbb{F}} F^{f,(s_f)}_{v_f} \qquad \big(v = (v_f)_f,\ s = (s_f)_f\big). \tag{1.5}$$

No entry of a product is declared. That $\{F_v\}_{v\in\mathcal{V}}$ is then a POVM on $Q$, block diagonal for the product sector map with the blocks of (1.5), follows for the exact numbers the factors declare: a tensor product of Hermitian positive semidefinite operators is Hermitian and positive semidefinite, $\sum_v \bigotimes_f F^f_{v_f} = \bigotimes_f \sum_{v_f} F^f_{v_f} = \mathbb{1}_Q$ by multilinearity, and the tensor product distributes over the direct sums $F^f_{v_f} = \bigoplus_{s_f} F^{f,(s_f)}_{v_f}$.

(DX4) *(the two forms of Alice's measurement slot)* In *POVM form* the slot holds $\{M_x\}_{x\in\mathcal{X}}$. In *ensemble form*, which exists in trusted-source mode only, it holds a family $\{N_x\}_{x\in\mathcal{X}}$ of block lists along $\mathbb{T}$, every block Hermitian and positive semidefinite, with

$$\sum_{x\in\mathcal{X}} N^{(t)}_x = \tau^{(t)}_A \quad \text{and} \quad \tau^{(t)}_A \neq 0 \qquad \text{for every } t\in\mathbb{T}, \tag{1.6}$$

the sum formed entrywise in the declared order of $\mathcal{X}$ and rounding nowhere, and the second condition a comparison of declared entries with the datum 0. An ensemble-form declaration *denotes* the POVM-form declaration with the same remaining slots and

$$M^{(t)}_x \coloneqq \big(\tau^{(t)}_A\big)^{-1/2} N^{(t)}_x \big(\tau^{(t)}_A\big)^{-1/2} + \frac{\operatorname{tr} N^{(t)}_x}{\operatorname{tr}\tau^{(t)}_A}\big(\mathbb{1}_{A_t} - P_t\big), \tag{1.7}$$

where $P_t$ is the projector onto the support of $\tau^{(t)}_A$ and the inverse square root is taken on that support and is zero off it. The operators (1.7) are defined and in general irrational; every statement made of an ensemble-form declaration is made of the declaration it denotes. For a POVM-form declaration in trusted-source mode put $N_x \coloneqq \tau_A^{1/2} M_x \tau_A^{1/2}$; for the declaration an ensemble form denotes, this returns the declared $N_x$.

(DX5) *(transcript protocol)* The announcement slot holds a *transcript protocol*, an ordered list of $L \geq 1$ entries, each a *message*, message $m$ carrying a sender $\pi_m \in \{\mathsf{A},\mathsf{B}\}$, a finite alphabet $\mathcal{C}_m$ and a total table

$a_m : \mathcal{O}_{\pi_m} \times \mathcal{C}_1 \times \cdots \times \mathcal{C}_{m-1} \to \mathcal{C}_m$. The joint announcement function is *derived* from the tables by composition,

$$\kappa_m(x,v) \coloneqq a_m\big(\pi_{\pi_m}(x,v),\, \kappa_1(x,v), \ldots, \kappa_{m-1}(x,v)\big), \qquad a(x,v) \coloneqq \big(\kappa_m(x,v)\big)_{m=1}^{L} \in \mathcal{K} \coloneqq \prod_{m=1}^{L} \tag{1.8}$$

and the acceptance-statistic map $\phi$, the sifting predicate $\varsigma$ of (1.3) and the holder's key map $r_{\mathsf{h}}$ of D(viii) of Definition 1.5 are maps on this $\mathcal{K}$. The two maps $a_A$, $a_B$ of the basic format are the transcript protocol with $L = 2$, $\pi = (\mathsf{A}, \mathsf{B})$, $\mathcal{C}_1 = \mathcal{C}_A$, $\mathcal{C}_2 = \mathcal{C}_B$, $a_1 = a_A$ and $a_2(o, \kappa_1) = a_B(o)$. A message is a function of its sender's readout and of the earlier messages of the same round, and of nothing else. No measurement of a round depends on a message of that round, the measurements being the fixed families of (DX3) and (DX4); processing across rounds is not a round message.

(DX6) *(party-scoped entries)* Each entry of $\mathfrak{L}$ carries the label of one measured factor, its *scope* $f_k$, and its operator $\hat{O}_k$ is a block list against that factor's own sector map, Hermitian block by block. Unless the entry is declared party-local it denotes the *lifted operator* $\mathbb{1} \otimes \hat{O}_k \otimes \mathbb{1}$ on $Q$, with $\hat{O}_k$ on the tensor factor $Q_{f_k}$, whose $s$-block is $\mathbb{1} \otimes \hat{O}_k^{(s_{f_k})} \otimes \mathbb{1}$, and it is read in the forms stated above for that lifted operator. With one measured factor the scope is that factor and the lifted operator is $\hat{O}_k$ itself. An entry may instead be declared party-local, in either source mode, carrying a real $c_k$; its reading is the third form of Definition 2.5.

*What this module decides of the general format* is Definition 1.3 for the operator data and D(i)–D(viii) of Definition 1.5 for the rest; that the family denoted by an ensemble form is a POVM is Theorem 5.1 (v).

**Definition 1.3** (Exact admissibility of declared operator data)**.** Declared operator data are binary64 numbers, each the dyadic rational it denotes and not an approximation of anything (Definition "Binary64 numbers, declared values and dyadic lattices" (foundations/def:binary64) (iii)), and every predicate this module decides on them is decided *exactly*, on those numbers. This definition lists those predicates, together with the one condition of the list that is a condition on the arithmetic that forms a sum rather than a predicate on an operator.

*The declared summation order.* The compilation map emits a total order on every declared alphabet and on $\mathbb{S}$: the label-to-index map $\mathsf{idx}$, which numbers the letters of each declared alphabet, and the sectors, consecutively from 0 (Definition 3.3, Theorem 3.4); the declared order of an alphabet is its $\mathsf{idx}$ order. For a finite family $\{a_j\}_{j \in J}$ of $n \times n$ matrices indexed by a

declared alphabet $J$ with idx order $j_1 \prec \cdots \prec j_{|J|}$, the sum in the declared order is accumulated as the computed sequential sum (Definition "Sums in a declared order and sums that round nowhere" (foundations/def:declared-order))

$$\Sigma_0 := 0, \qquad \Sigma_i := \Sigma_{i-1} + a_{j_i} \quad (1 \le i \le |J|), \qquad \Sigma := \Sigma_{|J|}, \tag{1.9}$$

formed left to right, entrywise, in binary64, with no re-ordering and no re-association; the trace of a matrix is likewise accumulated over its row index in increasing order, which is the order of the emitted array.

*The five conditions.* Let $\{a_j\}_{j\in J}$ be a finite family of block lists against a sector map $\mathfrak{s}$, one array $a_j^{(s)}$ on $\mathcal{H}(Q_s)$ per $s \in \mathbb{S}$. The family is exactly admissible for $\mathfrak{s}$ when

(T) *(sector map)* the sector map the supplied family carries is $\mathfrak{s}$, the map this module holds: the same block kind, the same block identifiers in the same order, the same dimensions, so that $a_j = \bigoplus_s a_j^{(s)}$ holds *by construction*. This is a type condition: it is decided, never measured, and an off-block entry is not small but inexpressible;

(S) *(the sum rounds nowhere)* writing $\Sigma^{(s)}$ for the total $\Sigma$ that (1.9) produces on the block family $\{a_j^{(s)}\}_j$: for each $s$, every one of the $|J|$ entrywise binary64 additions forming it is exact, so that the computed $\Sigma^{(s)}$ *is* the real sum $\sum_j a_j^{(s)}$;

(H) *(Hermiticity)* $a_j^{(s)} = a_j^{(s)\dagger}$ for every $(j, s)$;

(P) *(positivity)* $a_j^{(s)} \succeq 0$ for every $(j, s)$; the matrix being Hermitian by (H), this says that its spectrum is non-negative;

(C) *(completeness)* $\Sigma^{(s)} = \mathbb{1}_{Q_s}$ for every $s$.

Each of (H), (P), (C) is an *identity or an order relation between exactly represented numbers* and is decided as such: (H) and (C) entry by entry, (P) by exact symmetric elimination on the block with the zero-pivot row test (a negative pivot fails, and a zero pivot passes only when the rest of its row is zero), cubic in $d_s$. Together the five conditions are exact admissibility for $\mathfrak{s}$ with constant 1, that is with every completeness sum equal to the identity, in the sense of Definition "Exact admissibility of a declared operator family" (foundations/def:exact-admissibility), the declared order of $J$ being its idx order: (T), (H), (P) and (S) are the conditions of that definition bearing the same names, and (C), which reads the computed sum, is its completeness identity, because under (S) the computed sum $\Sigma^{(s)}$ is the exact sum $\sum_j a_j^{(s)}$ (Lemma "Exactness of a sum that rounds nowhere" (foundations/lem:exact-accumulation)).

*The same conditions on Alice's data, where there is one block and no sector map.* $A$ carries no sector decomposition, so a family on $A$ is one

array per index and (T) is empty for it. $\{M_x\}_{x\in\mathcal{X}}$ is *exactly admissible* when (S), (H), (P) and (C) hold of it on $A$ with $\mathbb{1}_A$ on the right of (C); the declared marginal $\tau_A$ satisfies its own three conditions when $\tau_A = \tau_A^\dagger$, $\tau_A \succeq 0$ and $\operatorname{tr}\tau_A = 1$ with the trace accumulated in the row order *without rounding*, the trace being a sum, read below as an identity of real numbers exactly as a completeness sum is.

*What is not on this list.* A strict inequality on a declared scalar, such as the condition of Definition 4.1 that every public-choice weight be positive, is a domain condition on a declared real number, decided by its sign and not by any of the five. A relation symbol, a declared tag and the type of a right-hand side are likewise decided and never measured (D(vi)). And nothing in this definition is a *norm*: no quantity here is compared against a tolerance window, and no defect, the entrywise size of the difference a condition names, is computed.

*The general format.* For a declaration in the general format of Definition 1.1 (DX1)–(DX6) the same conditions are read as follows. *Per measured factor:* for every $f \in \mathbb{F}$ the family $\{F^f_{v_f}\}_{v_f\in\mathcal{V}^f}$ is exactly admissible for the sector map of $f$ — (T) against that factor's map, and (S), (H), (P), (C) on that factor's blocks in that factor's declared order. No condition is decided on a product block (1.5), none being a datum; that the product family then satisfies (H), (P) and (C) on $Q$ as identities of real numbers is shown in (DX3). An entry's operator $\hat{O}_k$ is decided (T) and (H) against the map of its scope. *Alice's data, block by block:* with a factor tagged $A$, condition (T) is not empty for Alice's data: $\tau_A$, the operators of her measurement slot and a general $\Theta_k$ are block lists against that factor's map. In POVM form $\{M^{(t)}_x\}_x$ satisfies (S), (H), (P) and (C) on every $A_t$ with $\mathbb{1}_{A_t}$ on the right of (C); $\tau_A$ satisfies (H) and (P) block by block and $\sum_t \operatorname{tr}\tau^{(t)}_A = 1$, the traces accumulated in the row order and then over the blocks in their declared order, rounding nowhere. *Ensemble form:* the family $\{N^{(t)}_x\}_x$ satisfies (S), (H) and (P) on every $A_t$, and in place of (C) the identity $\Sigma^{(t)} = \tau^{(t)}_A$ of (1.6), entry by entry, between the declared-order sum and the declared block of $\tau_A$; and $\tau^{(t)}_A \neq 0$ for every $t$, a comparison of declared entries with the datum 0. These are all the conditions of the form: nothing is decided of the denoted $\{M_x\}$ of (1.7), its being a POVM a theorem (Theorem 5.1 (v)). At one Alice sector and one measured factor in POVM form these readings are the paragraphs above.

**Definition 1.5** (Target conditions of a well-formed declaration)**.** A declaration (1.1), *together with the label-to-index map* $\mathsf{idx}$ *emitted alongside it* (Definition 3.3), is a *well-formed declaration* when the following hold, every operator predicate below being decided *exactly* in the sense of Definition 1.3 and every combinatorial condition exactly as well:

D(i) all systems are finite-dimensional, all alphabets are finite and non-

empty, all declared maps are total, and *the sector map agrees*: there is one sector map $\mathfrak{s}$; every declared $Q$-operator is a block list against it, with the same block kind, the same identifiers in the same order and the same dimensions; and $d_Q = \sum_{s\in\mathbb{S}} d_s$. That much is condition (T) of Definition 1.3, stated once for the declaration. *And the order declared on the sectors is one order and not two*: the restriction of $\mathsf{idx}$ to $\mathbb{S}$ is the total order $\mathfrak{s}$ fixes. This last clause is a condition on the pair $(\mathfrak{D}, \mathsf{idx})$ and not on a slot of (1.1);

D(ii) the marginal slot holds $\tau_A$ with $\tau_A = \tau_A^\dagger$, $\tau_A \succeq 0$ and $\operatorname{tr}\tau_A = 1$, the trace summed in the row order and rounding nowhere — so that $\tau_A \in S_=(A)$ holds of the declared numbers, and a slot holding the zero matrix is inadmissible, its trace being $0$ — or the value $\mathsf{none}$;

D(iii) *(typing)* $\{F_v\}$ and every $\hat{O}_k$ are block lists against $\mathfrak{s}$ in the sense of Definition 1.1, so that $\Pi_s F_v \Pi_{s'} = 0$ and $\Pi_s \hat{O}_k \Pi_{s'} = 0$ for $s \neq s'$ hold *by construction* and are not conditions on data; and *(five predicates)* $M_x \succeq 0$ for each $x$ and $\{M_x\}$ complete on $A$; $F_v^{(s)} \succeq 0$ for each $(v, s)$ and $\{F_v^{(s)}\}_v$ complete on $Q_s$ for each $s$; and $\hat{O}_k^{(s)}$ Hermitian for each $(k, s)$, each decided exactly, each positivity predicate being the two decisions (H) and (P) of Definition 1.3 on the same block, and each completeness predicate carrying with it the condition (S) that its declared-order sum round nowhere. This is where Hermiticity of Alice's and Bob's emitted families enters the target conditions, and it is the only place it does;

D(iv) $r$ maps into $\mathcal{Z} \sqcup \{\perp\}$;

D(v) *sifting is announcement-determined*: there is $\varsigma : \mathcal{K} \to \{\text{pass}, \text{fail}\}$ with $r(x, v) = \perp \iff \varsigma(a(x, v)) = \text{fail}$;

D(vi) in trusted-source mode $\bowtie_k \in \{\preceq, =, \succeq\}$ and each entry carries one of the two declared forms of Definition 2.5, homogeneous with $c_k \in \mathbb{R}$ or general with $\Theta_k$ Hermitian on $A$; in untrusted-source mode $\bowtie_k \in \{\leq, =, \geq\}$ with $\Theta_k = c_k \in \mathbb{R}$. The relation symbol and the type of the right-hand side, which is what fixes the tag (Definition 2.5), are combinatorial and are read exactly; the Hermiticity of a general $\Theta_k$ is the exact predicate (H) on one matrix;

D(vii) the feasible set of Definition 2.1 is non-empty, $\mathcal{W} \neq \varnothing$;

D(viii) *party computability*: each party announces a function of its own readout ($a_A$ is defined on $\mathcal{O}_\mathsf{A}$ and $a_B$ on $\mathcal{O}_\mathsf{B}$, jointly (1.2)) and the declared holder computes the raw key from its own readout and the announcements: there is a total $r_\mathsf{h} : \mathcal{O}_\mathsf{h} \times \mathcal{K} \to \mathcal{Z}_\perp$ with

$$r(x, v) = r_\mathsf{h}\big(\pi_\mathsf{h}(x, v),\, a(x, v)\big) \qquad \text{for all } (x, v) \in \mathcal{X} \times \mathcal{V}. \tag{1.10}$$

The *default party partition* is $\pi_{\mathsf{A}}(x, v) = x$, $\pi_{\mathsf{B}}(x, v) = v$, with $A$ labelled $\mathsf{A}$ and every $Q_s$ labelled $\mathsf{B}$.

*The general format.* For a declaration in the general format of Definition 1.1 (DX1)–(DX6) the conditions read as follows, and in the basic format — one Alice sector, one measured factor, POVM form, two messages neither reading the other and no party-local entry — they are the eight above. D(i): the sector map is one factorised map (at most one factor tagged $A$, and then the first, at least one tagged $Q$, one block kind), every declared family on a measured factor and every entry's operator is a block list against the map of its factor, Alice's data against the factor tagged $A$, $d_Q = \prod_f \sum_{s_f} d_{f,s_f}$, and the restriction of $\mathsf{idx}$ to each factor's sectors is the order that factor's map fixes, the product order being derived and not declared. D(ii): $\tau_A$ is a block list along $\mathbb{T}$ with every block Hermitian and positive semidefinite and total trace 1, summed without rounding; in ensemble form additionally every block is non-zero. D(iii): the typing and the predicates of Definition 1.3 in their general-format reading, per measured factor and per Alice sector, the ensemble-form conditions in place of those on $\{M_x\}$ where that form is declared; ensemble form is refused in untrusted-source mode. D(vi): an entry carries its scope, which names a measured factor of the map, and one of the forms of Definition 2.5, the party-local form included, in either mode, with a real $c_k$; a general $\Theta_k$ is a block list along $\mathbb{T}$, Hermitian block by block. *And the relation symbol is read by form and not by mode alone*: an *untagged* entry carries $\bowtie_k \in \{\preceq, =, \succeq\}$ in trusted-source mode and $\bowtie_k \in \{\leq, =, \geq\}$ in untrusted-source mode, exactly as stated for D(vi), while an entry declared party-local carries $\bowtie_k \in \{\preceq, =, \succeq\}$ *in either mode*, its reading being the operator relation of the last line of (2.3) (Definition 2.5, the relation-symbol clause of the party-local form); where every row of that entry is $1 \times 1$ (every Alice block, and every sector block of every measured factor other than the entry's scope, of dimension one, $d_A = 1$ at one measured factor included) that relation is between $1 \times 1$ matrices and the three operator symbols are the three scalar symbols, which is the degenerate case. Where no entry is party-local, as in every declaration of the basic format, this clause is D(vi) as stated. D(viii): the announcement slot is a transcript protocol (DX5), every table $a_m$ being total on $\mathcal{O}_{\pi_m} \times \mathcal{C}_1 \times \cdots \times \mathcal{C}_{m-1}$, so that a message reads its sender's readout and earlier messages *by its signature*; the joint function $a$ is the composition (1.8), and (1.10) holds with it. D(iv), D(v) and D(vii) keep their wording, with $a$ the derived function and $\mathcal{W}$ the set of (2.3).

## 2 Feasible states and licensed constraints

**Definition 2.1** (Feasible set)**.** For a declaration $\mathfrak{D}$, the *feasible set* $\mathcal{W}$ is the set of $\omega$ presented as a sector-block list on $AQ$ — $\omega = \bigoplus_s \omega^{(s)}$ with $\omega^{(s)} \in \mathcal{L}(AQ_s)$, so that $(\mathbb{1}_A \otimes \Pi_s)\omega(\mathbb{1}_A \otimes \Pi_{s'}) = 0$ for $s \neq s'$ holds by the

presentation and is not a condition — such that

$$\omega^{(s)} \succeq 0 \;\; \forall s, \qquad \operatorname{tr}\omega = 1, \qquad \omega \models \mathfrak{D}, \qquad \omega \models \mathfrak{L}, \tag{2.1}$$

each read *exactly*, with $\operatorname{tr}\omega = \sum_s \operatorname{tr}\omega^{(s)}$ and with $\omega^{(s)} \succeq 0$ carrying, as in Definition 1.3, Hermiticity of the block. Here $\omega \models \mathfrak{D}$ is the marginal condition, $\omega_A = \tau_A$ in trusted-source mode, with $\omega_A = \sum_s \operatorname{tr}_{Q_s} \omega^{(s)}$, and no condition beyond $\operatorname{tr}\omega = 1$ in untrusted-source mode; and $\omega \models \mathfrak{L}$ reads

$$\begin{aligned} &\text{trusted:} \quad \operatorname{tr}_Q\big[(\mathbb{1}_A \otimes \hat{O}_k)\,\omega\big] = \sum_s \operatorname{tr}_{Q_s}\big[(\mathbb{1}_A \otimes \hat{O}_k^{(s)})\,\omega^{(s)}\big] \;\bowtie_k\; \Theta_k \quad \forall k; \\ &\text{untrusted:} \quad \operatorname{tr}\big[(\mathbb{1}_A \otimes \hat{O}_k)\,\omega\big] = \sum_s \operatorname{tr}\big[(\mathbb{1}_A \otimes \hat{O}_k^{(s)})\,\omega^{(s)}\big] \;\bowtie_k\; c_k \quad \forall k, \end{aligned} \tag{2.2}$$

where in the trusted line $\Theta_k$ is the *constructed* right-hand side $c_k\tau_A$ for an entry declared homogeneous and the *declared* Hermitian matrix for one declared general (Definition 2.5). The set carries no parameter, admits no tolerance window and relaxes no conjunct; in particular positivity is never relaxed.

*The general format.* For a declaration in the general format of Definition 1.1 (DX1)–(DX6), $\mathcal{W}$ is the set of $\omega$ presented as a list over the joint sectors, $\omega = \bigoplus_{(t,s)} \omega^{(t,s)}$ with $\omega^{(t,s)} \in \mathcal{L}(A_t Q_s)$ and every block positive semidefinite, so that block diagonality along $\mathbb{T}$ and along $\mathbb{S}$ both hold by the presentation, subject to the same conditions $\operatorname{tr}\omega = 1$, $\omega \models \mathfrak{D}$ and $\omega \models \mathfrak{L}$. Write $\hat{O}_k^{\uparrow(s)} \coloneqq \mathbb{1} \otimes \hat{O}_k^{(s_{f_k})} \otimes \mathbb{1}$ for the $s$-block of the lifted operator of an entry scoped to $f_k$ (DX6), and for a party-local entry scoped to $f$ split $s = (s_{f^c}, s_f)$, with $Q_{f^c,s_{f^c}}$ the product of the other factors' blocks. The conditions then read, block by block,

$$\begin{aligned} &\textstyle\sum_{(t,s)} \operatorname{tr}\omega^{(t,s)} = 1 && \text{(normalization)}; \\ &\textstyle\sum_s \operatorname{tr}_{Q_s}\omega^{(t,s)} = \tau_A^{(t)} \quad \forall t && \text{(marginal, trusted-source mode)}; \\ &\textstyle\sum_s \operatorname{tr}_{Q_s}\big[(\mathbb{1}_{A_t} \otimes \hat{O}_k^{\uparrow(s)})\,\omega^{(t,s)}\big] \bowtie_k \Theta_k^{(t)} \quad \forall t && \text{(trusted-source mode)}; \\ &\textstyle\sum_{(t,s)} \operatorname{tr}\big[(\mathbb{1}_{A_t} \otimes \hat{O}_k^{\uparrow(s)})\,\omega^{(t,s)}\big] \bowtie_k c_k && \text{(untrusted-source mode)}; \\ &\textstyle\sum_{s_f \in \mathbb{S}^f} \operatorname{tr}_{Q_{f,s_f}}\Big[\big(\mathbb{1} \otimes (\hat{O}_k^{(s_f)} - c_k\mathbb{1})\big)\,\omega^{(t,(s_{f^c},s_f))}\Big] \bowtie_k 0 \quad \forall (t, s_{f^c}) && \text{(party-local, either mode)}, \end{aligned} \tag{2.3}$$

the third line with $\Theta_k^{(t)} = c_k\tau_A^{(t)}$ for an entry declared homogeneous, the fourth for a scalar entry, and the last as an operator on $\mathcal{H}(A_t) \otimes \mathcal{H}(Q_{f^c,s_{f^c}})$. Each line is the corresponding operator condition on $AQ$: the marginal $\omega_A$, the entry value $\operatorname{tr}_Q[(\mathbb{1}_A \otimes \hat{O}_k)\omega]$, $\tau_A$ and every $\Theta_k$ are block diagonal along $\mathbb{T}$, the party-local value $\operatorname{tr}_{Q_f}[(\mathbb{1} \otimes (\hat{O}_k - c_k\mathbb{1}_{Q_f}))\,\omega]$ is block diagonal along the pairs $(t, s_{f^c})$, and an equality or a semidefinite inequality between operators block diagonal for one decomposition holds exactly when it holds

block by block. With $|\mathbb{T}| = 1$, one measured factor and no party-local entry, (2.3) is (2.1) with (2.2), line for line. With one measured factor the party-local line reads $\sum_s \mathrm{tr}_{Q_s}[(\mathbb{1}_{A_t} \otimes (\hat{O}_k^{(s)} - c_k\mathbb{1}))\omega^{(t,s)}] \bowtie_k 0$ on $\mathcal{H}(A_t)$, that is $\mathrm{tr}_Q[(\mathbb{1}_A \otimes \hat{O}_k)\omega] \bowtie_k c_k\,\omega_A$. The set is convex and compact: every line is an affine equality or an affine semidefinite or scalar inequality, and $\omega \succeq 0$ with $\mathrm{tr}\,\omega = 1$.

**Definition 2.5** (Licensed constraint list)**.** Each entry of $\mathfrak{L}$ carries its right-hand side in a declared form, and the form is part of the datum rather than a property measured of it (Definition 1.1). There are two *untagged* forms, called so because their tag is the type of their datum and needs no field of its own: a *homogeneous* entry declares a real scalar $c_k$, a *general* entry declares a Hermitian $\Theta_k$ on $A$; in untrusted-source mode only the scalar form exists. For these two forms the tag *is* the type of the right-hand side an entry carries, and it is read off that type: an entry whose right-hand-side slot holds a real scalar is homogeneous, and one whose slot holds a Hermitian matrix on $A$ is general. *That reading is a type decision*: it inspects which of two objects a slot holds, compares no numbers, reads no matrix entry and measures nothing. The third form, party-local, is declared by its tag and is not read off the type of the datum. $\mathfrak{L}$ is *licensed for the declared source mode* when every entry is in the form that mode admits:

(a) *trusted-source mode:* every entry is declared homogeneous, carrying $c_k \in \mathbb{R}$; the right-hand side it denotes is then *constructed* as $\Theta_k := c_k\tau_A$ from the declared $c_k$ and the declared $\tau_A$, and the entry of (2.2) reads $\mathrm{tr}_Q[(\mathbb{1}_A \otimes \hat{O}_k)\,\omega] \bowtie_k c_k\tau_A$;

(b) *untrusted-source mode:* every right-hand side is a real number $\Theta_k = c_k$ and the entry reads $\mathrm{tr}\Big[(\mathbb{1}_A \otimes \hat{O}_k)\,\omega\Big] \bowtie_k c_k$.

Licensing is therefore a *type* check on the tag, and it compares no numbers: what is inspected is which of the two forms an entry declares; no operator identity is tested against any number, and nothing here is decided at a window. *Only in cases (a) and (b) are the entries written* $(\hat{O}_k, \bowtie_k, c_k)$; a general entry is written $(\hat{O}_k, \bowtie_k, \Theta_k)$ and is the form the notation table records under $\Theta_k$. A general entry is admissible input, every structural claim of this contract being proved for it, but it is not licensed, so a list containing one carries no key-length claim (Theorem 9.4).

*The party-local form, a third licensed form.* In the general format of Definition 1.1 (DX6) an entry may be declared *party-local*: it carries its scope $f$, a block list $\hat{O}_k$ against the sector map of the measured factor $f$ and a real $c_k$, it is written $(\hat{O}_k, \bowtie_k, c_k; f)$, and its state-level reading, in either

source mode, is

$$\mathrm{tr}_{Q_f}\Big[\big(\mathbb{1}\otimes(\hat{O}_k - c_k\,\mathbb{1}_{Q_f})\big)\,\omega\Big] \;\bowtie_k\; 0 \qquad \text{as an operator on the systems of } \omega \text{ other than } Q_f, \tag{2.5}$$

those systems being $A$ and the measured factors $Q_{f'}$, $f' \neq f$; equivalently $\mathrm{tr}_{Q_f}[(\mathbb{1}\otimes\hat{O}_k)\,\omega] \bowtie_k c_k\,\mathrm{tr}_{Q_f}\,\omega$, the entry being homogeneous with respect to the marginal that is actually there. Block by block it is the last line of (2.3). The list is then *licensed* when every entry is either in the form the source mode admits, (a) or (b), read for its lifted operator, or is declared party-local. This is still a type check and compares no numbers: what is inspected is the declared tag. A party-local entry's right-hand-side slot holds a real scalar, as a homogeneous entry's does, so the type of the datum does not separate the two: an entry is party-local exactly when it carries that tag.

*The soundness condition of a party-local entry.* A party-local entry $(\hat{O}_k, \bowtie_k, c_k; f)$ meets the *soundness condition of a party-local entry* when every attack reaches $Q_f$ through a channel $\Lambda_f$ acting on that factor alone which carries the spectral certificate $\Lambda_f^\dagger(\hat{O}_k) \bowtie_k c_k\mathbb{1}$, that is, the adjoint of the channel applied to the entry's operator stands in the entry's relation to $c_k$ times the identity.

*What the tag changes, case by case.* With one measured factor and trusted-source mode, (2.5) reads $\mathrm{tr}_Q[(\mathbb{1}_A\otimes\hat{O}_k)\,\omega] \bowtie_k c_k\,\omega_A$, which under the marginal condition $\omega_A = \tau_A$ is the form (a) of the same $(\hat{O}_k, \bowtie_k, c_k)$. With one measured factor and $d_A = 1$, in either mode, it is the comparison of real numbers $\mathrm{tr}\big[\hat{O}_k\omega\big] \bowtie_k c_k\,\mathrm{tr}\,\omega = c_k$, the form (b). *With one measured factor, untrusted-source mode and $d_A > 1$ it is neither*: it is an operator inequality on $A$ against the marginal $\omega_A$ that is actually there, and the scalar entry (b) of the same $(\hat{O}_k, \bowtie_k, c_k)$ is its trace, implied by it and not equivalent to it. With more than one measured factor it implies, in either mode, the untagged reading of the same entry, which is obtained from it by the partial trace over the other measured factors (and over $A$ in untrusted-source mode), and it is in general not implied by it. So the tag never weakens an entry, and of the four one-factor cases, by source mode and by whether $d_A = 1$, it changes something only in untrusted-source mode at $d_A > 1$.

*Which relation symbols a tagged entry carries, in either mode.* An entry declared party-local carries $\bowtie_k \in \{\preceq, =, \succeq\}$ *in either source mode*, because (2.5) is a relation between *operators* in either mode: its left-hand side is an operator on the systems of $\omega$ other than $Q_{f_k}$, which are $A$ and the other measured factors. The scalar case is the degenerate one: where $d_{A_t}\prod_{f\neq f_k} d_{f,s_f} = 1$ for every row — every Alice block, and every sector block of every measured factor other than $f_k$, of dimension one, which with one measured factor means every Alice block of dimension one, whether $d_A = 1$ or $d_A = |\mathbb{T}| > 1$ — each row is a comparison of $1\times 1$ matrices, that is of real numbers, and the three operator symbols are then the three

scalar symbols $\leq$, $=$, $\geq$. With one measured factor and $d_A = 1$ there is one row, and the reading is the form (b) of the same $(\hat{O}_k, \bowtie_k, c_k)$, as the case-by-case paragraph states; otherwise there is one such comparison per row, and the reading relates to the untagged forms as that paragraph states for its case. *Nothing about the untagged forms changes*: an untagged entry carries the symbols its mode admits, and the operator symbols beside a tag are a property of the third form and of no other. No entry of a declaration in the basic format is tagged.

## 3 Specification, compilation map and typing

**Definition 3.1** (Protocol specification)**.** A *protocol specification* is a tuple

$$\mathfrak{S} = \big(\mathfrak{S}_{\mathrm{src}},\ \mathfrak{S}_{\mathrm{rec}},\ \mathfrak{R},\ \mathfrak{S}_{\mathrm{pub}},\ \mathfrak{S}_{\mathrm{con}}\big)$$

whose parts are:

*(a) Source part* $\mathfrak{S}_{\mathrm{src}}$. A source specification in the sense of Definition "Source specification" (transmitter/def:source-spec), presented to this module as the *transmitter artefact* in its declared mode: a finite non-empty alphabet $\mathcal{X}$ with its declared order, a finite-dimensional system $A$ with declared computational basis, a POVM $\{M_x\}_{x\in\mathcal{X}}$ on $A$ — itself possibly presented as a choice-weighted family (Definition 4.1) — and the specified signal ensemble $\{p_x, \rho_x\}_{x\in\mathcal{X}}$ on the transmitted system $A'$ (Assumption "Emission of the specified ensemble" (transmitter/ass:ens)). The mode is read off the artefact's marginal slot and is never a second datum. In trusted-source mode the slot holds $\tau_A \in S_=(A)$ — at $d_A = 1$ the slot the transmitter filled with [1] included — and at $d_A \geq 2$ the artefact additionally carries a source-replacement certificate $(\tau_A, \Psi, \{p_x, \rho_x^{\mathrm{v}}\}_{x\in\mathcal{X}})$ on $AA''$ in the sense of Definition "Source-replacement certificate" (transmitter/def:srcert), written against the virtual ensemble the source segment *starts from*, which lives on the virtual transmitted system $A''$, the source descriptor of $\Xi_1$ (Definition "Source segment and its interface conditions" (transmitter/def:source-segment) (S1), (S3)). When $L_{\mathrm{s}} = 0$ that system and that ensemble are the specified ones: $A'' = A'$ and $\rho_x^{\mathrm{v}} = \rho_x$. In untrusted-source mode the slot holds none and there is no certificate.

*(b) Receiver part* $\mathfrak{S}_{\mathrm{rec}}$. A finite-dimensional physical receiver system with a POVM $\{F_w^{\mathrm{ph}}\}_{w\in\mathcal{V}^{\mathrm{ph}}}$, possibly presented as a choice-weighted family; and, optionally, a public choice whose weight depends on the private setting of the measuring apparatus — a stochastic matrix $P(f \mid t)$ with the setting map $t : \mathcal{V} \to \mathcal{T}$ it is read through — to be applied by this module *after* the reduction chain, in the setting-dependent form of Definition 4.1.

*(c) Reduction chain* $\mathfrak{R} = (\mathfrak{R}_{\mathrm{src}}, \mathfrak{R}_{\mathrm{rec}})$. A reduction chain in the sense of Definition "Reduction chain" (reductions/def:chain), presented to this module as a record: an ordered *pair* of ordered lists of typed links, the *source*

*segment* $\mathfrak{R}_{\mathrm{src}} = (\Xi_1, \ldots, \Xi_{L_s})$ with $L_s \geq 0$, which is a source segment in the sense of Definition "Source segment and its interface conditions" (transmitter/def:source-segment) — its links act on the transmitted system $A'$ and replace Alice's signal ensemble — and the *receiver segment* $\mathfrak{R}_{\mathrm{rec}} = (\Lambda_1, \ldots, \Lambda_L)$ with $L \geq 1$, whose links act on the receiver system and replace Bob's measurement. What this module reads of a link, stated in its own words, is this: each link of either segment carries a type tag, source and target system descriptors, a status tag, its channel as data where the link type has one, the constraint entries it transports, and a licence slot. A receiver-stage link in the sense of Definition "Typed link" (reductions/def:link) carries more — its reachable-set statement and its carried operator pair — and this module reads none of it. The last link of $\mathfrak{R}_{\mathrm{rec}}$ has target descriptor $(Q, \mathbb{S}, (d_s)_s)$ with the sector projectors $\Pi_s$, and target POVM $\{F_v\}_{v \in \mathcal{V}}$. The two segments are separate lists because they have different types; the pair is composed source segment first. The interface conditions on a chain that this module decides on the record are those of Definition 7.1.

*(d) Public part* $\mathfrak{S}_{\mathrm{pub}}$. A party partition $\mathfrak{P}$ (readouts $\pi_{\mathsf{A}}, \pi_{\mathsf{B}}$ onto $\mathcal{O}_{\mathsf{A}}, \mathcal{O}_{\mathsf{B}}$, and the party labels), announcement maps $a_A : \mathcal{O}_{\mathsf{A}} \to \mathcal{C}_A$ and $a_B : \mathcal{O}_{\mathsf{B}} \to \mathcal{C}_B$, a sifting predicate $\varsigma : \mathcal{K} \to \{\mathrm{pass}, \mathrm{fail}\}$, a key rule $r^0_{\mathsf{h}} : \mathcal{O}_{\mathsf{h}} \times \mathcal{K} \to \mathcal{Z}$, a holder $\mathsf{h} \in \{\mathsf{A}, \mathsf{B}\}$, a finite acceptance alphabet $\widehat{\mathcal{C}}$ and a map $\phi : \mathcal{K} \to \widehat{\mathcal{C}}$. All maps are total.

*(e) Constraint part* $\mathfrak{S}_{\mathrm{con}}$. A finite, possibly empty, list of entries. Each $\hat{O}_k$ is presented as a block list on $Q$ together with the sector map it is written against, which is carried with it as for every other supplied $Q$-operator family, and is Hermitian block by block. Each entry carries a right-hand-side datum of one of two types, and the type fixes the entry's tag: homogeneous with a real $c_k$, or general with a Hermitian $\Theta_k$ on $A$, in trusted-source mode; a real $c_k$ in untrusted-source mode (Definition 2.5). The constraint part also carries the optional attack-set licence record and an optional feasibility witness $\omega^\star$, likewise a block list with its own map. The map each $\hat{O}_k$ carries is compared with the emitted $\mathfrak{s}$ by check V1, an entry declared here being on the same footing as one transported by a link. The witness's map is not compared there: $\omega^\star$ is a block list on $AQ$ and not a $Q$-operator, it is no slot of (1.1), and branch (c) of check V7 is where it is required to be a block list against $\mathfrak{s}$ (Definition 6.1).

*The general format.* A specification to be compiled into the general format of Definition 1.1 (DX1)–(DX6) has the same five parts, read as follows. At one Alice sector, one measured factor, POVM form, two messages neither of which reads the other and no party-local entry these readings are (a)–(e) above, part for part on every component a derivation reads: at those values every such object named below is the object of the same name there and nothing else is presented. On the party labels of the tensor factors of each $\mathcal{H}(Q_s)$, which no derivation of this module reads and which the consuming

engine's laboratory premise reads as naming the parties' laboratories, (DX1) restricts the party partition of (d). The party set is unchanged: it is $\{\mathsf{A}, \mathsf{B}\}$, with the two readout maps of $\mathfrak{P}$. What is new is that several measured factors may carry a party label and that the factor tagged $A$ carries one too (DX1).

*(a′) Source part.* The artefact's marginal slot may hold a block list $(\tau_A^{(t)})_{t\in\mathbb{T}}$ along the Alice sectors the artefact declares, and Alice's measurement arrives in one of the two forms of (DX4): the POVM slot, or, in trusted-source mode only, the ensemble slot holding the block lists $\{N_x\}_{x\in\mathcal{X}}$ of (1.6). The source-replacement certificate may accordingly be declared in pure form, in block form, or as an ensemble presentation, all three in the sense of Definition "Source-replacement certificate" (transmitter/def:srcert). Which form a certificate is in is read off the artefact and is never a second datum, exactly as the mode is. That a form may be *presented* is the transmitter's rule and is stated here. Which forms this module *admits*, and what gate (G2) decides of each form, is Definition 9.1's and is stated once there.

*(b′) Receiver part.* $\mathfrak{S}_{\mathrm{rec}}$ is an ordered list, one entry per measured factor $f \in \mathbb{F}$ and in the order of $\mathbb{F}$: a party label; a finite-dimensional physical receiver system with a POVM $\{F_{w_f}^{\mathrm{ph},f}\}_{w_f\in\mathcal{V}^{\mathrm{ph},f}}$, possibly presented as a choice-weighted family; and, optionally, a setting-dependent public choice at that factor — a stochastic matrix $P_f(\,\cdot \mid \cdot\,)$ with the setting map on that factor's alphabet it is read through — to be applied after that factor's segment, in the setting-dependent form of Definition 4.1. No entry of the list carries an operator on another factor's system, and no product of physical families is presented: the product is the rule (1.5), formed by no one.

*(c′) Reduction chain.* $\mathfrak{R} = (\mathfrak{R}_{\mathrm{src}}, \mathfrak{R}_{\mathrm{rec}})$ with $\mathfrak{R}_{\mathrm{src}}$ as in (c) and $\mathfrak{R}_{\mathrm{rec}} = (\mathfrak{R}_{\mathrm{rec}}^f)_{f\in\mathbb{F}}$ one receiver segment per measured factor, each $\mathfrak{R}_{\mathrm{rec}}^f = (\Lambda_1^f, \ldots, \Lambda_{L_f}^f)$, $L_f \geq 1$, a list of receiver-stage links whose source descriptor is factor $f$'s physical receiver system and whose last target descriptor is $(Q_f, \mathbb{S}^f, (d_{f,s_f})_{s_f})$ with target POVM $\{F_{v_f}^f\}_{v_f\in\mathcal{V}^f}$. What this module reads of a link is what (c) lists and nothing more, of each segment separately: the segments are separate lists because their links act on different systems, and no link of one is composable with a link of another. Where the factors' data are the ordered list of the factors' artefacts that the reductions module emits, factor $f$'s receiver segment and target family are that factor's own, and the product rule of Theorem "The emission of several measuring parties" (reductions/thm:emitted-product) is emitted in place of a joint family; that list is what the factor list of (DX1) is built from, in its order. The interface conditions on a chain that this module decides on the record are those of Definition 7.1, per factor for the receiver segments.

*(d′) Public part.* In place of $a_A$ and $a_B$ the public part declares the transcript of (DX5): the number $L \geq 1$ of messages and, for each $m$, its sender $\pi_m$, its finite alphabet $\mathcal{C}_m$ and its total table $a_m : \mathcal{O}_{\pi_m} \times \mathcal{C}_1 \times \cdots \times$

$\mathcal{C}_{m-1} \to \mathcal{C}_m$. The sifting predicate $\varsigma$, the key rule $r^0_{\mathsf{h}}$, the holder $\mathsf{h}$, the acceptance alphabet $\widehat{\mathcal{C}}$ and $\phi$ are declared on the derived $\mathcal{K} = \prod_m \mathcal{C}_m$ of (1.8), which is a set determined by the tables and declared by no one. All tables are total. The party partition $\mathfrak{P}$ is as in (d), with a party label per measured factor and, where there is one, for the factor tagged $A$.

*(e′) Constraint part.* Each entry presents, in addition to what (e) lists, its scope: the label of one measured factor of the chain; and $\hat{O}_k$ is a block list on $Q_{f_k}$ together with the sector map of that factor, Hermitian block by block. Its right-hand-side datum is of one of the three forms of Definition 2.5: a real $c_k$ tagged homogeneous, a Hermitian $\Theta_k$ tagged general and then itself a block list along $\mathbb{T}$, or a real $c_k$ tagged party-local. The arriving tag is compared with the type of that datum at check V6, which also decides that the scope names a measured factor of the emitted map. An optional witness $\omega^\star$ is a block list over the joint sectors with its own map, compared at branch (c) of V7 and nowhere else, for the reason (e) gives.

**Definition 3.2** (Well-formed specification)**.** A protocol specification $\mathfrak{S}$ (Definition 3.1) is a *well-formed specification* when its declared data are as declared: (i) all alphabets are finite and non-empty; (ii) all systems are finite-dimensional with the declared dimensions; (iii) all tabulated maps are total on their stated domains; (iv) all declared POVM families are exactly admissible: every family other than the physical receiver family $\{F^{\mathrm{ph}}_w\}_{w\in\mathcal{V}^{\mathrm{ph}}}$ as presented, block by block in the sense of Definition 1.3; and the physical receiver family as presented (in the general format, each measured factor's, on that factor's physical receiver system), on its physical receiver space read as one block, condition (T) being empty for it as Definition 1.3 reads Alice's data: where it is presented directly, (S), (H), (P) and (C) of that definition hold of it in the declared order of its alphabet; where it is presented as a choice-weighted family (Definition 4.1), its conditional families are exactly admissible, block by block against the map they carry, and condition (S) holds of its own declared-order sum. A choice-weighted family whose conditional families carry a sector map is a block list against that map, and reading it as that block list or as one block decides the same (S), (H), (P) and (C), its off-block entries being exact zeros; (v) all public-choice weights are strictly positive — a domain condition on a declared scalar (Definition 4.1) — and sum to one; (vi) every setting-dependent weighting, at whichever measured system or measured factor it is declared, is a stochastic matrix, its entries non-negative and its rows summing to one exactly (the setting-dependent form of Definition 4.1); (vii) every constraint operator is a block list against the upstream module's sector map and Hermitian block by block, and every entry carries its declared right-hand-side tag.

**Definition 3.3** (The compilation map)**.** The *compilation map* is the map $\mathfrak{S} \mapsto \mathfrak{D}$ that fills (1.1) as follows.

- *Alice's system and measurement.* $A$, $d_A$ and $\{M_x\}$ from $\mathfrak{S}_{\mathrm{src}}$, weighted by Definition 4.1 where a choice family is presented.

- *Alice's marginal.* $\tau_A$ is what the transmitter artefact's marginal slot holds: in trusted-source mode the matrix in the slot — at $d_A = 1$ the value [1] the transmitter filled the slot with, the unique element of $S_{=}(A)$ — and $\tau_A = \mathsf{none}$ where the slot is empty. The mode is read off that slot and off nothing else. A certificate the artefact carries is consumed by gate (G2) of Definition 9.1.

- *The measured system and its family.* $(Q, \mathbb{S}, (d_s)_s)$ and $\{F_v\}$ from the last link of $\mathfrak{R}_{\mathrm{rec}}$. Where $\mathfrak{S}_{\mathrm{rec}}$ declares a setting-dependent public choice, the family is weighted after the chain by the setting-dependent form of Definition 4.1: the emitted family is then $N_{v,f} = P(f \,|\, t(v))\, F_v$ over the support of $P$, the emitted outcome alphabet is that support, and the setting map $t$ through which $P$ is read is the class map of the joint alphabet of the chain's producer where the chain was emitted by the reductions module (Definition "Joint alphabet, classes and the joint family of a private setting" (reductions/def:joint-family) (a)), and is declared with the family otherwise.

- *The public maps.* $\mathfrak{P}$, $a_A$, $a_B$, $\phi$, $\mathsf{h}$ verbatim from $\mathfrak{S}_{\mathrm{pub}}$, declared on the emitted alphabets.

- *The constraint list.* $\mathfrak{L}$ is the concatenation of $\mathfrak{S}_{\mathrm{con}}$'s entries with the entries transported by the links of $\mathfrak{R}_{\mathrm{src}}$ and then those of $\mathfrak{R}_{\mathrm{rec}}$, each segment in its own order, each entry with the datum its right-hand-side slot holds and with the tag that datum's type determines (Definition 2.5).

- *The raw-key map.*

$$r(x,v) := \begin{cases} r^0_{\mathsf{h}}\big(\pi_{\mathsf{h}}(x,v),\, a(x,v)\big), & \varsigma(a(x,v)) = \text{pass}, \\ \perp, & \varsigma(a(x,v)) = \text{fail}, \end{cases} \qquad \text{with } d_Z := |\mathcal{Z}|. \tag{3.1}$$

*The index map.* Alongside $\mathfrak{D}$ the compiler emits the *index map* $\mathsf{idx}$, the label-to-index map: a choice, for each of $\mathcal{X}$, $\mathcal{V}$, $\mathbb{S}$, $\mathcal{O}_{\mathsf{A}}$, $\mathcal{O}_{\mathsf{B}}$, $\mathcal{C}_A$, $\mathcal{C}_B$, $\widehat{\mathcal{C}}$ and $\mathcal{Z}_{\perp}$, of a bijection onto an initial segment of $\mathbb{Z}_{\geq 0}$; every emitted list, matrix and function table is indexed through $\mathsf{idx}$, and every semantic label is carried as metadata only. Its component on $\mathbb{S}$ is the total order the sector map $\mathfrak{s}$ already fixes, and not a second choice: one order on the sectors, required by the last clause of D(i) (Definition 1.5) and decided by V1. And $\mathsf{idx}$ is what fixes the declared summation order of Definition 1.3 on every alphabet it indexes; it indexes no basis of a matrix, a trace being accumulated in the row order of the emitted array.

*The binding rule.* On the remaining alphabets the choice is the compiler's, under one binding rule. Where a supplied family arrives with a declared order on the alphabet that indexes it — as the target POVM of the last link of $\mathfrak{R}_{\mathrm{rec}}$ does, its declarer having established its own completeness sum in that order — the compiler takes *that* order as $\mathsf{idx}$'s component there. So the order in which condition (S) of Definition 1.3 is decided on a supplied family here is the order that family was declared in, and no family is refused here for an ordering its declarer never chose. Where no order arrives with an alphabet the choice is free, and (S) is decided in the order the compiler chose and emits. This rule prescribes nothing to the declarer of a family: it constrains only which bijection this module picks.

Where a choice-weighted presentation is resolved, this same rule settles the order the decision on each conditional family is read in, and adds nothing to it. The family the weighting produces is indexed by one alphabet — $\mathcal{X}$ where the presentation is Alice's, the physical receiver alphabet where it is a measuring party's, which in Band A is the emitted $\mathcal{V}$ — and where a declared order arrives with that alphabet the rule above takes *it*, here as everywhere: $\mathcal{X}$ arrives ordered with the source part (Definition 3.1 (a)), and the physical alphabet with the order the fibre sum of Definition 7.1 (C4) is accumulated in. Condition (S) on the conditional family of a choice value is then read in the *restriction of that order* to the elements that choice value contributes, which is a total order on $\mathcal{U}_t$ however the record listed it; the two conditions are independent, only the order restricting. Where no order arrives with that alphabet the free choice is made as Definition 4.1 says, by the record's own listing, and it is emitted. No declarer's order is overridden and no presentation is refused for an order. Where the setting-dependent weighting is applied, the emitted alphabet is the support of $P$ and its order is the product order of Definition 4.1: $v$ in its declarer's declared order and, within each $v$, $f$ in the declared order of the choice alphabet, the pairs off the support omitted; that is the order (S) is decided in on the emitted family and the order $\mathsf{idx}$ emits on it.

*The general format.* For a specification presented as Definition 3.1's (a′)–(e′) the map fills the general format of Definition 1.1 (DX1)–(DX6) as follows. In the basic format — one Alice sector, one measured factor, POVM form, two messages neither reading the other and no party-local entry — every clause below is the clause of the same subject above, slot for slot, and the map emits what those clauses emit.

*The factor list* (DX1) is assembled and not declared as a whole: the factor tagged $A$ is the Alice sector map the transmitter artefact declares, where it declares one, and it is the first of the list; where the artefact declares none, $\mathbb{T} = \{t_0\}$ and no $A$ factor is emitted. The factors tagged $Q$ are the entries of $\mathfrak{S}_{\mathrm{rec}}$ in their declared order, each carrying that entry's party label and the sector map of the last target descriptor of that entry's receiver segment, which Definition 7.1 (C1) decides per factor. The one block kind

is the kind those maps carry, and the factors' agreement in it is decided by the same four-component comparison V1 makes, taken on the factor list (Definition 6.1, V1). The product sector map, its order, the product alphabet and the joint sectors are derived from that list by (DX1) and (DX3) and are declared, formed and emitted by no one: what is emitted is the factor list and the rules.

*Alice's slots.* $\tau_A$ is what the artefact's marginal slot holds, the block list along $\mathbb{T}$ included; Alice's measurement slot is the artefact's, in the form the artefact declares — the POVM slot or the ensemble slot of (DX4) — weighted by Definition 4.1 in that form where a choice family is presented. The mode is read off the marginal slot as above.

*Per measured factor.* $(Q_f, \mathbb{S}^f, (d_{f,s_f})_{s_f})$ and $\{F^f_{v_f}\}_{v_f \in \mathcal{V}^f}$ from the last link of that factor's receiver segment and, where that factor's entry of $\mathfrak{S}_{\mathrm{rec}}$ declares a setting-dependent public choice, weighted after that segment by the setting-dependent form of Definition 4.1, so that factor $f$'s emitted family is $P_f(\,\cdot \mid \cdot\,)$ times that factor's delivered family over the support of $P_f$, and factor $f$'s emitted alphabet $\mathcal{V}^f$ is that support in its product order. Each factor's weighting is applied to its own family and to nothing else. No product of two factors' families is formed by this map, whether emitted or handed on; so the subject of the representability decision of Definition 4.1 is, as in the basic format, the products *that definition's* two constructions form, and no product block (1.5) is a product this map forms.

*The public part and the raw key.* $\mathfrak{P}$, $\phi$, $\mathsf{h}$ verbatim from $\mathfrak{S}_{\mathrm{pub}}$ and the transcript verbatim: the sender, the alphabet and the table of each message, declared on the emitted alphabets. The joint announcement function $a$ is not a slot: it is the composition (1.8) of those tables, and $r$ is (3.1) with *that* $a$, which is what Theorem 5.1 (iv), (vi) is stated of. $\mathfrak{L}$ is the concatenation above, each entry carrying in addition its scope, and each with the datum its right-hand-side slot holds and the tag that datum's type determines; for a party-local entry that is the tag field it arrives with beside a real scalar (Definition 2.5).

*The index map in the general format.* $\mathsf{idx}$ has one component per factor alphabet $\mathcal{V}^f$ and one per factor sector set $\mathbb{S}^f$, and, where an $A$ factor is emitted, one on $\mathbb{T}$, which is the total order that factor's own map fixes. Its components on $\mathcal{V}$ and on $\mathbb{S}$ are the product orders (DX1) and (DX3) derive from the factors' orders and are not further choices, as its component on $\mathbb{S}$ above is not; the joint sectors are ordered with $t$ slowest by (DX1). The binding rule above is applied per factor: where a factor's target family arrives with a declared order on its alphabet, the compiler takes that order there. So condition (S) of Definition 1.3 is decided per factor, in that factor's arriving order, and per Alice sector on Alice's data; no sum over a product alphabet is accumulated by this map or by its consumer.

**Theorem 3.4** (Typing of an emitted declaration)**.** Let $\mathfrak{S}$ be a well-formed

specification (Definition 3.2) satisfying Assumption 10.1, and let $\mathfrak{D}$ be a declaration that the compilation map of Definition 3.3 *emits*, that is, one for which no check of Definition 6.1 refused. Then $\mathfrak{D}$, together with the index map $\mathsf{idx}$ emitted alongside it, satisfies the target conditions D(i), D(ii), D(iv) and D(vi) of Definition 1.5, and $\mathsf{idx}$ is a bijection on each alphabet it indexes, so that every emitted operator list is totally ordered and its index is recoverable from its semantic label.

**Definition 3.6** (Declared statistics functional)**.** For a declaration $\mathfrak{D}$ and $\omega \in \mathcal{L}(AQ)$ put

$$P_{\mathfrak{D}}[\omega](x,v) := \operatorname{tr}\big[(M_x \otimes F_v)\,\omega\big], \qquad \mathbf{\Phi}[\omega](\hat{c}) := \sum_{(x,v):\,\phi(a(x,v))=\hat{c}} P_{\mathfrak{D}}[\omega](x,v). \tag{3.2}$$

Both are defined by trace formulas and are therefore defined, and affine, on all of $\mathcal{L}(AQ)$. The feasible set $\mathcal{W}$ of Definition 2.1 is contained in $\mathrm{Pos}(AQ)$, and on it the values are a probability distribution (Theorem 3.7 (iv), and (iv′) in the general format).

**Theorem 3.7** (Exact measurement families and the statistics functional)**.** Let $\mathfrak{S}$ be a well-formed specification (Definition 3.2) satisfying Assumption 10.1, and let $\mathfrak{D}$ be emitted by Definition 3.3 in the basic format. Then:

(i) $\{M_x\}_{x\in\mathcal{X}}$ is a POVM on $A$ and, for each $s \in \mathbb{S}$, the block family $\{F_v^{(s)}\}_{v\in\mathcal{V}}$ is a POVM on $Q_s$ — each predicate holding *of the declared numbers*, decided exactly — and each of the completeness sums $\sum_x M_x$ and $\sum_v F_v^{(s)}$, formed in the declared order, rounds nowhere, so that each computed sum *is* the sum it is written as (Definition 1.3);

(ii) every $F_v$ and every $\hat{O}_k$ is sector-block-diagonal *by construction*: it is emitted as the list $\big(F_v^{(s)}\big)_s$, respectively $\big(\hat{O}_k^{(s)}\big)_s$, and the operator it denotes is $\bigoplus_s F_v^{(s)}$. The two formulations of block-diagonality remain equivalent — for an operator $F$ on $Q$, $\Pi_s F \Pi_{s'} = 0$ for all $s \neq s'$ if and only if $F = \bigoplus_s \Pi_s F \Pi_s$ — and that equivalence is what licenses the *type*: a list of blocks denotes exactly the operators the vanishing condition picks out, and no other. It is not checked, because the condition has no data to be checked on;

(iii) hence D(iii) holds: its typing half by (ii) and check V1; its first four predicates — those on $\{M_x\}$ and on each block family $\{F_v^{(s)}\}_v$ — by (i); and its fifth, Hermiticity of $\hat{O}_k^{(s)}$ for each $(k,s)$, by the last clause of check V3 (Definition 6.1), which decides it exactly on the emitted constraint entries and which (i) does not state; and

(iv) $P_{\mathfrak{D}}[\omega]$ of (3.2) and $\mathbf{\Phi}[\omega]$ are affine in $\omega$, $\mathbf{\Phi}[\omega]$ depends on $\omega$ only through $P_{\mathfrak{D}}[\omega]$, and the fibres of $\phi \circ a$ partition $\mathcal{X} \times \mathcal{V}$, so $\mathbf{\Phi}[\omega]$ is the pushforward of $P_{\mathfrak{D}}[\omega]$ and inherits its total mass. For every block list $\omega$ with $\omega^{(s)} \succeq 0$ for each $s$ and $\operatorname{tr} \omega = 1$ — in particular for every $\omega \in \mathcal{W}$ — $P_{\mathfrak{D}}[\omega]$ is a probability distribution on $\mathcal{X} \times \mathcal{V}$ and $\mathbf{\Phi}[\omega]$ one on $\widehat{\mathcal{C}}$.

*The general format.* Let $\mathfrak{D}$ instead be emitted by the general-format clauses of Definition 3.3, in the general format of Definition 1.1 (DX1)–(DX6). Then (i)–(iv) hold with the following readings, of which the four above are the case of the basic format.

(i′) for every measured factor $f$ and every $s_f \in \mathbb{S}^f$ the block family $\{F_{v_f}^{f,(s_f)}\}_{v_f \in \mathcal{V}^f}$ is a POVM on $Q_{f,s_f}$; and for every Alice sector $t$, in POVM form $\{M_x^{(t)}\}_{x \in \mathcal{X}}$ is a POVM on $A_t$, and in ensemble form every $N_x^{(t)}$ is Hermitian and positive semidefinite, with $\sum_x N_x^{(t)} = \tau_A^{(t)}$ and $\tau_A^{(t)} \neq 0$ — each predicate holding of the declared numbers, decided exactly, and each of these sums, formed in its declared order, rounding nowhere. The product family (1.5) is a POVM on $Q$ and, in ensemble form, the family (1.7) is a POVM on $A$, both as identities of real numbers, and neither is formed, decided or summed;

(ii′) every $F_{v_f}^f$ is a block list against factor $f$'s sector map and every $\hat{O}_k$ one against the sector map of its scope, and $\tau_A$, every operator of Alice's measurement slot and every general $\Theta_k$ is a block list along $\mathbb{T}$, each block diagonal by construction as in (ii); and the product family is block diagonal for the product sector map, $\Pi_s F_v \Pi_{s'} = 0$ for $s \neq s'$, with the $s$-block of (1.5);

(iii′) hence D(iii) holds in its general-format reading (Definition 1.5); and

(iv′) (iv) holds for every list $\omega$ over the joint sectors with every block positive semidefinite and $\operatorname{tr} \omega = 1$ — in particular for every $\omega \in \mathcal{W}$ — with $F_v$ the product (1.5) and, in ensemble form, $M_x$ the operator (1.7) of the declaration $\mathfrak{D}$ denotes: $P_{\mathfrak{D}}[\omega]$ is a probability distribution on $\mathcal{X} \times \mathcal{V}$ and $\mathbf{\Phi}[\omega]$ one on $\widehat{\mathcal{C}}$.

# 4 Public choices as measurement weights

**Definition 4.1** (Choice-weighted families)**.** Let $S$ be a finite-dimensional system, $\mathcal{T}_{\mathrm{pub}}$ a finite non-empty *public-choice alphabet* with declared weights $q_t > 0$, $\sum_{t \in \mathcal{T}_{\mathrm{pub}}} q_t = 1$, and for each $t \in \mathcal{T}_{\mathrm{pub}}$ a POVM $\{N_{u|t}\}_{u \in \mathcal{U}_t}$ on $S$. The *choice-weighted family* is indexed by the disjoint union $\mathcal{V} := \bigsqcup_{t \in \mathcal{T}_{\mathrm{pub}}} \mathcal{U}_t \times \{t\}$ and defined by

$$N_{u,t} := q_t \, N_{u|t}, \qquad (u,t) \in \mathcal{V}. \tag{4.1}$$

Several independent choices compose by iterating (4.1); the resulting weight is the product of the individual weights. *What is declared, and therefore what is decided, is the innermost family and each weight list*: an iterated presentation declares $\{N_{u|t}\}_u$ for each value of each choice together with one weight list per choice, the decision on the conditional families is taken on those, and the intermediate families are this module's own products, which inherit admissibility by Lemma 4.2 (i) at each step and carry no second declared condition; the representability decision is taken on every product the map forms, the intermediate ones included. This is the *setting-independent form*: the weight of a choice value is one declared number, the same in every round and whatever else either party's apparatus did.

*A weighting that depends on a station's setting.* Let $\{F_v\}_{v\in\mathcal{V}}$ be a POVM on $S$ over a finite alphabet $\mathcal{V}$ with a declared order — the target family of the last link of $\mathfrak{R}_{\mathrm{rec}}$, as Definition 3.3 takes it — and let $t : \mathcal{V} \to \mathcal{T}$ be a declared total map onto a finite non-empty *setting alphabet*, the *setting component* of the outcome: the value of the private per-round setting of the measuring apparatus that the outcome $v$ records. Where the chain was emitted by the reductions module, $t$ is the class map of that module's joint alphabet (Definition "Joint alphabet, classes and the joint family of a private setting" (reductions/def:joint-family) (a): $\mathcal{V} = \mathcal{Y} \times \mathcal{T}$ with classes $\mathcal{V}_t$); where the family is declared directly, $t$ is declared with it, the singleton $\mathcal{T} = \{*\}$ being the case of no setting. Let $\mathcal{F}$ be a finite non-empty *choice alphabet* with a declared order and let $P(f\,|\,t)$, $(t, f) \in \mathcal{T}\times\mathcal{F}$, be a *stochastic matrix*: $P(f\,|\,t) \geq 0$ and $\sum_{f\in\mathcal{F}} P(f\,|\,t) = 1$ for every $t \in \mathcal{T}$. The *support of the weighting* is

$$\mathcal{V}_P \coloneqq \big\{(v, f) \in \mathcal{V} \times \mathcal{F} :\ P(f\,|\,t(v)) > 0\big\}, \tag{4.2}$$

ordered by the product order — $v$ in the declared order of $\mathcal{V}$ and, within each $v$, $f$ in the declared order of $\mathcal{F}$, the pairs off the support omitted — and the choice-weighted family in its *setting-dependent form* is

$$N_{v,f} \coloneqq P(f\,|\,t(v))\, F_v, \qquad (v, f) \in \mathcal{V}_P. \tag{4.3}$$

The setting-independent form is the case $P(f\,|\,t) = q_f$ for every $t$, in which (4.3) is (4.1) with $\mathcal{T}_{\mathrm{pub}} = \mathcal{F}$ and $N_{v|f} = F_v$ for every $f$ — one conditional family, weighted by a choice that does not see the setting; conversely, (4.1) with conditional families that differ with the choice is not of the form (4.3), and the two forms are two constructions overlapping in that case. *In the overlap the two constructions do not agree on an order, and the order is a property of the presentation and not of the family*: the alphabet the first form produces is $\bigsqcup_f \mathcal{U}_f \times \{f\}$, listed choice-major, while the support (4.2) of the second is listed outcome-major, so the two are transposes of one another and one family declared through the two records has its condition (S) decided in two different orders. (S) is order-dependent by construction (Definition 1.3), and the two orders are not reconciled: what fixes the order is the record that

was declared, and a declarer that declares the same family both ways must expect the decision its own record's order gives. *$P$ is read through the setting and through nothing else.* It is declared on $\mathcal{T} \times \mathcal{F}$ and reaches an outcome only through $t(v)$. Zero entries of $P$ are admitted, and the support is what they are for: a pair $(v, f)$ with $P(f \,|\, t(v)) = 0$ is not an outcome, is not emitted and indexes nothing, so every emitted element of (4.3) carries a strictly positive weight, exactly as $q_t > 0$ requires of every element of (4.1). Which choice values a given setting admits is therefore declared by the zero pattern of $P$ and not by a zero-weight outcome. The form is applied by the compiler *after* the reduction chain, to the family the last link delivered (Definition 3.3).

*The reading of each declared condition.* The weights are binary64 numbers and the conditional families binary64 matrices. So $\sum_t q_t = 1$ is decided *exactly*, with the sum accumulated in the declared order of $\mathcal{T}_{\mathrm{pub}}$ — the order in which the declared public-choice record lists $\mathcal{T}_{\mathrm{pub}}$, fixed there because idx does not index $\mathcal{T}_{\mathrm{pub}}$ (Definition 3.3) — and required to round nowhere; and "$\{N_{u|t}\}_u$ is a POVM on $S$" is exact admissibility in the sense of Definition 1.3, per block where $S$ carries sectors: (H), (P) and (C) per block; (S) on the completeness sum $\sum_{u \in \mathcal{U}_t} N_{u|t}$, accumulated in the order stated below and required to round nowhere; and (T) as the agreement — in block kind, in block identifiers, in their order and in their dimensions — of the maps *the conditional families of one presentation themselves carry*: there is one such map for the presentation, and the family the weighting produces is a block list against *it*. On a system carrying no sector decomposition, Alice's $A$, (T) is empty. This condition is decided of each $t \in \mathcal{T}_{\mathrm{pub}}$ separately, on the declared numbers.

*In which order (S) is read on a conditional family.* The family the weighting produces is indexed by one alphabet, and that alphabet carries one order: $\mathcal{X}$ where the presentation is Alice's, which arrives with its declared order (Definition 3.1 (a)); the physical receiver alphabet where it is a measuring party's, which is the order the fibre sum of Definition 7.1 (C4) is accumulated in and, in Band A, is the emitted alphabet's own. Condition (S) on the conditional family $\{N_{u|t}\}_u$ is decided in the *restriction of that order* to the elements the choice value $t$ contributes — a total order on $\mathcal{U}_t$ whatever the presentation listed, so no compatibility of the record's listing with it is required and none is decided. Where a declared order arrives with that alphabet the compiler takes it, by the binding rule of Definition 3.3, to which this definition adds nothing; where none arrives, the choice the compiler makes and emits is the record's own listing, $t$ in the order it lists $\mathcal{T}_{\mathrm{pub}}$ in and, within each $t$, $u$ in the order it lists $\mathcal{U}_t$ in. *No presentation is refused for an order*: what is fixed here is which order a decision is read in, not what a declarer may declare.

*The remaining declared conditions, at the same reading.* The strict inequality $q_t > 0$ is a domain condition on a declared real scalar, decided by

its sign, and Definition 1.3 carries no strict-inequality row because none is wanted — $q_t \neq 0$ is what Lemma 4.2 (ii) needs, and a negative $q_t$ would reverse the sign of an element of the emitted family, which no public choice of known bias can mean. For the setting-dependent form the same reading applies to $P$: each entry is a binary64 number, $P(f \mid t) \geq 0$ is decided by sign, each row sum $\sum_f P(f \mid t) = 1$ is decided exactly in the declared order of $\mathcal{F}$ and required to round nowhere, the support (4.2) is decided by the sign of each entry, and "$\{F_v\}$ is a POVM on $S$" is the exact admissibility the chain's (C3) already decided of the family the last link delivered. *This form has no second conditional family to decide*: the choice value is drawn after the measurement, $\{F_v\}$ is the whole of what the decision on the conditional families has as its subject here, and (C3) is where it was taken.

*What this module constructs, and the two emission decisions.* The products $q_t N_{u|t}$ of (4.1) and $P(f \mid t(v))F_v$ of (4.3) are formed *exactly* on the declared dyadic numbers. A product of two binary64 numbers need not be one, so the compiler decides, before emitting anything, that every entry of every $N_{u,t}$, respectively of every $N_{v,f}$, is a binary64 number — on every product this map forms, whether the product is emitted or handed on — and that the emitted family satisfies Definition 1.3 — (H), (P), (C) and, on its own declared-order sum over $\mathcal{V}$, respectively over $\mathcal{V}_P$ in its product order, (S). These are the two *emission decisions*.

*The lattice of the products.* Let the arriving entries be integer multiples of $2^{-k}$ of modulus at most 1, and let each $q_t$ be an integer multiple of $2^{-k_q}$; then every entry of $q_t N_{u|t}$ is an integer multiple of $2^{-(k+k_q)}$ of modulus at most 1, so the constructed family lies on a lattice $k_q$ steps finer than the one it was handed and is exactly binary64 whenever $k + k_q \leq 53$. Where a $q_t$ is not a power of two the finer lattice can leave binary64 altogether: the decimal literal 0.9 is an integer multiple of $2^{-53}$ and of no coarser one, so $k + k_q > 53$ for every $k \geq 1$ and the exact product of a declared entry by it need not be a binary64 number. The same arithmetic governs $P(f \mid t(v))F_v$ with the exponent $k_P$ of the entries of $P$ in place of $k_q$: a $P$ whose non-zero entries are dyadic with small exponent — $\frac{1}{2}$, $\frac{1}{4}$, 1 — composes with any family the chain delivers at $k \leq 53 - k_P$, and a decimal literal in $P$ can leave binary64 exactly as a decimal $q_t$ can.

*The general format.* In the general format of Definition 1.1 (DX1)–(DX6) both forms above are read *on the system of the party that declares them*, and nothing else of this definition changes. At one measured factor, in POVM form and with no factor tagged $A$, the readings below are the text above, sentence for sentence.

*Which system $S$ is.* For a presentation of Alice's it is $A$, and (T) is *not empty where a factor is tagged $A$*: the conditional families are then block lists along $\mathbb{T}$ and (T) is the four-component agreement of the maps they carry with that factor's map, as on any other sectored system. For a presentation of a measuring party's it is that party's *own* measured factor: one

presentation per factor, decided against that factor's own map; *a presentation indexed by a product alphabet has no declared system to be a family on* — the product is the rule (1.5) and is no datum (Definition 3.3, the general format). So the decision "$\{N_{u|t}\}_u$ is a POVM on $S$" and the products this module forms are per factor, and the representability decision keeps exactly the subject it has: the products *these two constructions* form, and no product block of (1.5).

*The ensemble form of Alice's slot.* Where Alice's measurement slot is declared in ensemble form (DX4), a choice-weighted presentation of it declares, for each $t \in \mathcal{T}_{\mathrm{pub}}$, a family $\{N_{x|t}\}_{x\in\mathcal{X}}$ of block lists along $\mathbb{T}$ with every block Hermitian and positive semidefinite, and *in place of the POVM condition* the ensemble condition of (1.6): for every Alice sector $t'$, $\sum_x N_{x|t}^{(t')} = \tau_A^{(t')}$, the sum accumulated in the order stated above and rounding nowhere. That condition is decided *of each $t$ separately and on the declared numbers*, with (H), (P) per block, that identity in place of (C), and (T) as above. *Why the construction is the same construction.* With $N_{x,t} \coloneqq q_t N_{x|t}$ of (4.1), the exact real numbers the data denote satisfy

$$\sum_{(x,t)\in\mathcal{V}} N_{x,t}^{(t')} = \sum_{t\in\mathcal{T}_{\mathrm{pub}}} q_t \sum_{x\in\mathcal{X}} N_{x|t}^{(t')} = \Big(\sum_{t\in\mathcal{T}_{\mathrm{pub}}} q_t\Big)\tau_A^{(t')} = \tau_A^{(t')} \tag{4.4}$$

for every $t'$, by the decision just taken and by $\sum_t q_t = 1$; so a choice-weighted presentation in ensemble form whose conditional ensembles are admitted denotes a weighted family with the ensemble identity of (1.6). *That is a consistency statement and not a substitute for the decision on the emitted family*: the emitted family's own (S), (H), (P) and its identity with the declared $\tau_A$ are decided at check V3, on the emitted blocks and in the emitted order, and neither condition implies the other: (4.4) is an identity of real numbers and the emitted decision is about the additions that compute it. In untrusted-source mode there is no ensemble form and this paragraph has no subject.

*The setting-dependent form, per measured factor.* It is applied by the compiler after *that factor's* receiver segment, to the family that factor's last link delivered, with that factor's own setting map, support (4.2) and product order; the setting component read is a component of *that factor's* outcome, the private per-round setting of the station that measures that factor, whichever party's label the factor carries. This form has no second conditional family here either, per factor: (C3) of Definition 7.1, read for that factor, is where the delivered family was decided.

**Lemma 4.2** (Properties of the choice-weighted families)**.** With the notation of Definition 4.1, and reading its conditions exactly:

(i) $\{N_{u,t}\}_{(u,t)\in\mathcal{V}}$ is a POVM on $S$; and so is $\{N_{v,f}\}_{(v,f)\in\mathcal{V}_P}$ of the setting-dependent form, over the support of $P$;

(ii) if $S = Q$ carries the sector decomposition and every $N_{u|t}$ is sector-block-diagonal, then so is every $N_{u,t}$, and conversely (for $q_t > 0$); equivalently, on block lists the weighting acts blockwise, $N^{(s)}_{u,t} = q_t\, N^{(s)}_{u|t}$, so a weighted family is a block list against the same sector map as the conditional families it was built from. The same holds of the setting-dependent form: every $N_{v,f}$ is a scalar multiple of the $F_v$ it was built from, $N^{(s)}_{v,f} = P(f \mid t(v))\, F^{(s)}_v$ block by block, and $N_{v,f}$ is sector-block-diagonal iff $F_v$ is, the weight being strictly positive on the support;

(iii) *(the choice marginal, for the setting-independent form)* for every $\omega \in \mathcal{L}(S)$ with $\operatorname{tr}\omega = 1$ and every $t \in \mathcal{T}_{\mathrm{pub}}$,

$$\sum_{u \in \mathcal{U}_t} \operatorname{tr}[N_{u,t}\,\omega] = q_t, \tag{4.5}$$

independently of $\omega$; and for any complementary system $R$ and any $\omega \in \mathcal{L}(RS)$ of unit trace likewise $\sum_u \operatorname{tr}[(\mathbb{1}_R \otimes N_{u,t})\,\omega] = q_t$, the operator being written with its factors in the declared order of the tensor factors. For a choice declared on Bob's side, $S = Q$ and $R = A$, this reads $\sum_u \operatorname{tr}[(\mathbb{1}_A \otimes N_{u,t})\omega] = q_t$; for a choice declared on Alice's side, $S = A$ and $R = Q$, it reads $\sum_u \operatorname{tr}[(N_{u,t} \otimes \mathbb{1}_Q)\omega] = q_t$. *This part is asserted of the setting-independent form and of it alone*;

(iv) *(what is asserted of the setting-dependent form instead)* forgetting the choice value returns the family the chain delivered,

$$\sum_{f\colon (v,f) \in \mathcal{V}_P} N_{v,f} = F_v \qquad \text{for every } v \in \mathcal{V}, \tag{4.6}$$

an exact operator identity on $S$; and the marginal of the choice value is the declared matrix applied to the *setting marginal* of the outcome statistics,

$$\sum_{v\colon (v,f) \in \mathcal{V}_P} \operatorname{tr}[N_{v,f}\,\omega] = \sum_{t \in \mathcal{T}} P(f \mid t) \sum_{v\colon t(v) = t} \operatorname{tr}[F_v\,\omega] \qquad \text{for every } \omega \in \mathcal{L}(S), \tag{4.7}$$

which depends on $\omega$ only through the *setting marginal* $m_t(\omega) \coloneqq \operatorname{tr}[\Pi_t \omega]$, $\Pi_t \coloneqq \sum_{v\colon t(v)=t} F_v$, with $\Pi_t \succeq 0$ and $\sum_t \Pi_t = \mathbb{1}_S$. If $P(f \mid t) = q_f$ for every $t$ it equals $q_f$ for every unit-trace $\omega$ — the setting-independent case, and (4.5) in the form the first construction gives it. If every class sum $\Pi_t$ is a multiple $\lambda_t \mathbb{1}_S$ of the identity — as for the identity link, where the emitted family is the physical joint family — it is state-independent as well, with value $\sum_t P(f \mid t)\lambda_t$. Otherwise it is, in general, a function of the state, and *no identity of the form* (4.5) *is asserted for this form.*

**Theorem 4.4 (Encoding of public choices in the measurement operators).** Let a per-round public choice of known bias — a basis choice, a choice between test and generation rounds, a probe setting — be declared on either party's side with alphabet $\mathcal{T}_{\mathrm{pub}}$, weights $(q_t)$ and conditional POVMs $\{N_{u|t}\}$, under Assumption 10.4. Then the compiler forms the choice-weighted family (4.1), and (iv) below holds. Where that family is the emitted one — the presentation is Alice's, or it is a measuring party's and the last link of that party's receiver segment delivers the weighted family itself, on its own alphabet, as its target POVM, as on the Band A route (Definition 8.1 (b)) — the compiler emits it, and (i)–(iii) hold as well:

(i) the emitted family is a POVM and preserves sector block-diagonality (Lemma 4.2 (i),(ii)), so the encoding cannot break D(iii) — and it is emitted only after the two decisions of Definition 4.1, so that what is emitted is binary64 data whose own predicates D(iii) is then decided on (Theorem 3.7 (i), and (i′) in the general format);

(ii) the encoding is *exact*: for every state, the marginal probability of the choice value $t$ computed from the emitted declaration equals the declared weight $q_t$ (Lemma 4.2 (iii));

(iii) the choice value is a component of the joint outcome, hence enters every downstream quantity only through $P_{\mathfrak{D}}[\omega]$ and $\mathbf{\Phi}[\omega]$ of Definition 3.6; no factor $q_t$ is emitted anywhere outside a POVM element; and

(iv) the alternative encoding — declaring the conditional families $\{N_{u|t}\}$ together with a separate state-independent fraction $q_t$ to be applied outside the operators — is never emitted, and a specification requesting it is refused.

Where the presentation is a measuring party's and the last link of that party's receiver segment delivers any other family, the weighted family is the physical receiver family handed to the chain (Definition 3.1 (b)) and is not emitted: it is a POVM on the physical receiver space, and for every unit-trace state there the marginal probability of $t$ computed from it is $q_t$ (Lemma 4.2 (i), (iii)); no factor $q_t$ is emitted anywhere outside a POVM element, and (iv) holds. The emitted family is that link's target POVM, whose own conditions are (C3)'s and check V3's (Definition 7.1), and nothing here asserts (ii) or (iii) of it; the declared test-statistics model is the output-state model, as (vii) says of the setting-dependent form. Let, further, a public choice $f \in \mathcal{F}$ be declared whose weight depends on the private setting of the apparatus that measures the measured system, as a stochastic matrix $P(f\,|\,t)$ read through the setting map $t : \mathcal{V} \to \mathcal{T}$ of the family $\{F_v\}_{v\in\mathcal{V}}$ the last link of $\mathfrak{R}_{\mathrm{rec}}$ delivered — the setting-dependent form of Definition 4.1; in the general format, at a measured factor, read through that factor's setting map on the family the last link of that factor's receiver

segment delivered, with $S$ that factor's system — under Assumption 10.4. Then the compiler applies it *after* the reduction chain, emits the family (4.3), $N_{v,f} = P(f \mid t(v))F_v$ over the support $\mathcal{V}_P$, and:

(v) the emitted family is a POVM and preserves sector block-diagonality (Lemma 4.2 (i),(ii)), every element being a scalar multiple of an element the chain delivered, so the encoding cannot break D(iii); and it is emitted only after the two decisions of Definition 4.1 on the products $P(f \mid t(v))F_v$, so that what is emitted is binary64 data whose own predicates D(iii) is then decided on (Theorem 3.7 (i), and (i′) in the general format);

(vi) *(soundness: the relabelling commutes with the chain)* for every channel $\Lambda$ into $S$ and every family $\{G_v\}_{v\in\mathcal{V}}$ on its input space with $\Lambda^\dagger(F_v) = G_v$ for every $v$,

$$\Lambda^\dagger\big(N_{v,f}\big) = P(f \mid t(v))\, G_v \qquad \text{for every } (v,f) \in \mathcal{V}_P : \tag{4.8}$$

the weighted target and the identically weighted input family stand in the same relation under the *same* channel, with the same target space, the same sector map, the same constraint entries transported along the chain and every pull-back unchanged. In this module's setting $G$ is the $\gamma$-coarse-grained physical family of Definition 7.1 (C4), $G_v = \sum_{w\in\gamma^{-1}(v)} F^{\mathrm{ph}}_w$, so the emitted family and the physical family weighted by $P(f \mid t(\gamma(w)))$ satisfy the pullback identity (7.1) with the outcome correspondence $(w,f) \mapsto (\gamma(w), f)$ restricted to the support, wherever (C4) is decidable, and are related as (C4)'s families are wherever it is not. This is Lemma "Linearity: post-processing, private settings and mixtures" (reductions/lem:linear) (i) for the stochastic matrix from $\mathcal{V}$ to $\mathcal{V}\times\mathcal{F}$ with entries $\delta_{vv'}P(f \mid t(v'))$;

(vii) *(what is not asserted)* the exactness clause (ii) is asserted for the setting-independent form and for it alone. For this form the marginal of $f$ is the quantity (4.7) of Lemma 4.2 (iv), a function of the state through the setting marginal, equal to the declared $q_f$ for every state when $P(f \mid t) = q_f$ and state-dependent in general otherwise; nothing here asserts otherwise, no state-independent fraction is emitted, and the declared test-statistics model is the output-state model. What (iv) says of the fixed-fraction encoding holds of this form unchanged;

(viii) *(refusals)* a weighting presented through anything but the setting map — indexed by $\mathcal{V}$, by the device outcome, or by any component of $v$ other than $t(v)$ — is refused; a $P$ with a negative entry or a row whose declared-order sum is not exactly 1 is refused; and the two emission decisions of Definition 4.1 refuse, for a product that is not representable or with the predicate that failed. Each refusal emits no declaration.

# 5 Sifting, key computability and feasibility

**Theorem 5.1** (Announcement locality, sifting and party computability)**.** Let $\mathfrak{S}$ be a well-formed specification (Definition 3.2) satisfying Assumption 10.1, so that in particular $r^0_{\mathsf{h}}$ maps into $\mathcal{Z}$ (Assumption 10.1 (ii)), and let $\mathfrak{D}$ be emitted by Definition 3.3. Then:

(i) *(announcement locality)* $a_A$ is a function of $\pi_{\mathsf{A}}(x, v)$ alone and $a_B$ of $\pi_{\mathsf{B}}(x, v)$ alone, and the joint announcement is (1.2);

(ii) *(D(v))* the specification's $\varsigma$ — consumed, and not itself a slot of (1.1) — witnesses announcement-determined sifting: $r(x, v) = \perp \iff \varsigma(a(x, v)) =$ fail;

(iii) *(D(viii))* $r$ is party-computable: the total map

$$r_{\mathsf{h}}(o, \kappa) := \begin{cases} r^0_{\mathsf{h}}(o, \kappa), & \varsigma(\kappa) = \text{pass}, \\ \perp, & \varsigma(\kappa) = \text{fail}, \end{cases} \qquad (o, \kappa) \in \mathcal{O}_{\mathsf{h}} \times \mathcal{K}, \tag{5.1}$$

satisfies (1.10).

(iv) *(transcript protocols)* if the announcement slot holds a transcript protocol (Definition 1.1 (DX5)), then for every $m$ the $m$-th component of $a(x, v)$ is a function of $\pi_{\pi_m}(x, v)$ and of the components before it, and of nothing else; $a$ is total; and (ii) and (iii) hold with this $a$. For $L = 2$, $\pi = (\mathsf{A}, \mathsf{B})$ and $a_2$ independent of $\kappa_1$ this is (i);

(v) *(ensemble form)* if Alice's measurement slot is declared in ensemble form and the conditions of Definition 1.3 for that form hold, then the operators (1.7) satisfy $M^{(t)}_x \succeq 0$ and $\sum_x M^{(t)}_x = \mathbb{1}_{A_t}$ for every $t$, and $(\tau^{(t)}_A)^{1/2} M^{(t)}_x (\tau^{(t)}_A)^{1/2} = N^{(t)}_x$: the denoted declaration satisfies the POVM clause of D(iii) on $A$ as an identity of real numbers, and $\operatorname{tr}[M_x \tau_A] = \sum_t \operatorname{tr} N^{(t)}_x$.

(vi) *(the compiled general format)* if $\mathfrak{S}$ is presented as Definition 3.1 (a′)–(e′) and $\mathfrak{D}$ is emitted by the general-format clauses of Definition 3.3, then: the announcement slot of $\mathfrak{D}$ holds the transcript $\mathfrak{S}_{\text{pub}}$ declares, message for message; the joint announcement function of $\mathfrak{D}$ is (1.8) evaluated at *those* tables and is a slot of nothing; and Alice's measurement slot holds the form the transmitter artefact declares. Consequently (iv) applies to $\mathfrak{D}$, hence (ii) and (iii) hold of it, and (v) applies to it exactly where that form is the ensemble form.

Moreover, under the default party partition these read: $a_A : \mathcal{X} \to \mathcal{C}_A$, $a_B : \mathcal{V} \to \mathcal{C}_B$, and $r(x, v) = r_{\mathsf{A}}(x, a(x, v))$ for $\mathsf{h} = \mathsf{A}$, respectively $r(x, v) = r_{\mathsf{B}}(v, a(x, v))$ for $\mathsf{h} = \mathsf{B}$.

**Theorem 5.5** (Feasibility by an explicit or a verified candidate)**.** Let $\mathfrak{D}$ be a candidate declaration satisfying D(i)–D(iii), and let $\mathcal{W}$ be as in Definition 2.1. (The hypothesis is on the conditions alone: nothing below uses that $\mathfrak{D}$ was emitted by Definition 3.3, and Theorem 6.2 applies this theorem to candidates.)

(i) If $\mathfrak{L} = \varnothing$ then $\mathcal{W} \neq \varnothing$. Explicitly, the block list

$$\omega^{\star(s)} = \begin{cases} \tau_A \otimes \dfrac{\mathbb{1}_{Q_s}}{d_Q}, & \text{trusted-source mode,} \\ \dfrac{\mathbb{1}_A}{d_A} \otimes \dfrac{\mathbb{1}_{Q_s}}{d_Q}, & \text{untrusted-source mode,} \end{cases} \qquad s \in \mathbb{S}, \tag{5.2}$$

which denotes $\tau_A \otimes \mathbb{1}_Q/d_Q$, respectively $\mathbb{1}_A/d_A \otimes \mathbb{1}_Q/d_Q$, lies in $\mathcal{W}$.

(ii) If $\mathfrak{L} \neq \varnothing$ and $\mathfrak{D}$ satisfies D(vi) as well, which makes each constraint leg of (2.1) a well-posed comparison, then $\mathcal{W} \neq \varnothing$ holds if and only if some block list $\omega$ satisfies the conditions of (2.1). The compiler decides those conditions *exactly*, by finitely many arithmetic operations and per block, on at most two candidates and in a fixed order: first the *default candidate* (5.2), which satisfies every conjunct of (2.1) except possibly the constraint legs (by the proof of (i)) and which the compiler forms in order to test and never emits; and, if that candidate fails a constraint leg, a *supplied* witness $\omega^\star$, consumed and verified and never emitted. It refuses exactly when the default candidate fails and either no witness is supplied or the supplied one fails.

*The general format.* For a candidate declaration in the general format of Definition 1.1 (DX1)–(DX6) satisfying D(i)–D(iii), and D(vi) for part (ii), with $\mathcal{W}$ the set of (2.3), both parts hold with the joint sectors in place of the sectors. In (i) the default candidate is the block list over the joint sectors

$$\omega^{\star(t,s)} = \begin{cases} \tau_A^{(t)} \otimes \dfrac{\mathbb{1}_{Q_s}}{d_Q}, & \text{trusted-source mode,} \\ \dfrac{\mathbb{1}_{A_t}}{d_A} \otimes \dfrac{\mathbb{1}_{Q_s}}{d_Q}, & \text{untrusted-source mode,} \end{cases} \qquad (t,s) \in \mathbb{T} \times \mathbb{S}, \tag{5.3}$$

and in (ii) the verification is the per-joint-block one listed in the proof, which forms no array larger than one joint block. At $|\mathbb{T}| = 1$ and one measured factor (5.3) is (5.2) and the two lists of decisions coincide item for item.

# 6 The emitted validator

**Definition 6.1** (The emitted validator)**.** Given a candidate declaration (1.1) with its emitted index map, the checks are as follows. *Every operator condition below is decided exactly*, in the sense of Definition 1.3, and every combinatorial condition exactly as well.

**V1 (finiteness, totality, and the sector map)**
every declared alphabet is finite and non-empty and every declared map is tabulated on all of its stated domain; every declared system carries a finite dimension; $d_Q = \sum_{s\in\mathbb{S}} d_s$; the restriction of $\mathsf{idx}$ to $\mathbb{S}$ is the total order the sector map fixes; *and the sector maps agree*: there is one emitted map $\mathfrak{s}$, and the map carried by each supplied $Q$-operator family is compared against it in *block kind*, in the *list of block identifiers*, in their *order* and in their *dimensions*. A disagreement on any of the four is refused, naming the first position at which the lists differ. The families compared are exactly the $Q$-operator families of (1.1), which is the list D(i) quantifies over and no wider one: the emitted family $\{F_v\}$, which the last link of $\mathfrak{R}_{\mathrm{rec}}$ delivers as its target POVM against the map the chain's producer wrote it against, and each constraint entry of $\mathfrak{L}$, declared in $\mathfrak{S}_{\mathrm{con}}$ or transported by a link of either segment. Those two slots are the whole of (1.1) that carries a $Q$-operator (Definition 1.1), so the list is exhaustive by construction rather than by enumeration; the objects outside it are compared where they belong. The entries of $\mathfrak{S}_{\mathrm{con}}$ are compared as well as the transported ones: Definition 3.3 fills $\mathfrak{L}$ from both sources, Definition 3.1 (e) has each declared $\hat{O}_k$ carry its own sector map, and an entry written against a different map is the error this check exists for, whichever source it arrived from. The comparison is combinatorial — it compares labels and integers, never matrix entries — and it is the whole of the structural verification the emitted declaration receives: block-diagonality is carried by the type (Theorem 3.7 (ii)), so the one structural error that can still arise is two parties disagreeing about what the sectors are, and V1 is the check for it.

**V2 (marginal slot)**
the slot holds $\mathsf{none}$, or a matrix $\tau_A$ on $A$ — one block, $A$ carrying no sector decomposition — satisfying $\tau_A = \tau_A^\dagger$, $\tau_A \succeq 0$ and $\operatorname{tr}\tau_A = 1$, each decided exactly, and with the trace accumulated over the row index verified to round nowhere; a failure of the last is refused.

**V3 (operators, block by block)**
$M_x = M_x^\dagger$ and $M_x \succeq 0$ for each $x \in \mathcal{X}$, and $\sum_x M_x = \mathbb{1}_A$ with the sum, in the emitted index order, verified to round nowhere, a failure of which is refused; $F_v^{(s)} = F_v^{(s)\dagger}$ and $F_v^{(s)} \succeq 0$ for each $(v, s) \in \mathcal{V} \times \mathbb{S}$, and, for each $s$, $\sum_v F_v^{(s)} = \mathbb{1}_{Q_s}$ with that sum verified to round nowhere, a failure being refused with the sector and the outcome index at which the first inexact addition occurred; and $\hat{O}_k^{(s)} = \hat{O}_k^{(s)\dagger}$ for each $(k, s)$. Every one of these is decided exactly. There is no off-block condition: an emitted $Q$-operator is a block list, so there is no $\Pi_s F_v \Pi_{s'}$ or $\Pi_s \hat{O}_k \Pi_{s'}$ to evaluate and nothing that could be non-zero.

**V4 (key alphabet)**
every tabulated value of $r$ lies in $\mathcal{Z} \sqcup \{\bot\}$.

**V5 (sifting consistency)**
for all $(x, v)$ and $(x', v')$ with $a(x, v) = a(x', v')$: $r(x, v) = \bot \iff r(x', v') = \bot$.

**V6 (constraint typing, including the tag)**
when the marginal slot is filled: $\bowtie_k \in \{\preceq, =, \succeq\}$, and the entry's right-hand side is of one of the two admitted types, which is what fixes its tag (Definition 2.5): homogeneous, whose datum is of real scalar type, or general, whose datum is a matrix on $A$ with $\Theta_k = \Theta_k^\dagger$ decided exactly. When the slot holds none: $\bowtie_k \in \{\leq, =, \geq\}$ and the right-hand side is of real scalar type. A real scalar right-hand side is accepted in trusted-source mode at every $d_A$, and is the licensed form there (Definition 2.5). The relation symbol, the type of the right-hand side and the tag that type fixes are compared exactly; an entry whose right-hand-side slot holds neither admitted object, or a complex-typed right-hand side in untrusted-source mode, is refused by type and not measured. The check also reads an object (1.1) does not carry, the tag field of an *arriving* entry, in $\mathfrak{S}_{\mathrm{con}}$ or transported by a link: where an arriving entry carries an explicit tag field, V6 compares that field with the type of the datum it names, and a disagreement is refused. An entry of (1.1) in either of the two forms above is the triple $(\hat{O}_k, \bowtie_k, \mathrm{rhs}_k)$ and holds no tag field independent of the type of its datum, the emitted tag being read off that type (Definition 2.5); so on a candidate declaration this clause has no subject, refuses nothing, and is not among the conjuncts D(vi) is paired with (Theorem 6.2, proof). An entry carrying no tag field is not refused: its tag is the type of its datum and is read off it, which is what lets an entry declared as a real constant, carrying no tag field, be read here.

**V7 (feasibility; one-sided, three branches)**
V7 passes in exactly three cases:

(a) $\mathfrak{L} = \varnothing$; or

(b) $\mathfrak{L} \neq \varnothing$ and the default candidate (5.2), which the compiler forms and does not emit, passes the per-block verification of Theorem 5.5 (ii); or

(c) $\mathfrak{L} \neq \varnothing$, the default candidate fails, and a witness $\omega^\star$ is supplied *as a block list against* $\mathfrak{s}$ — the map it carries compared with $\mathfrak{s}$ in block kind, in identifiers, in their order and in their dimensions, exactly as V1 compares the map of a declared $Q$-operator family — and passes that same verification. This is the one place a

witness's sector map is compared, and it is here rather than at V1 because $\omega^\star$ is a block list on $AQ$, is no slot of (1.1), and is quantified over by no target condition (Definition 1.5).

Otherwise V7 fails and the candidate is refused, the refusal naming, where a witness was supplied and rejected on its map, the first position at which the two lists differ. The verification in (b) and (c) is the $|\mathbb{S}|$ block positivity decisions — Hermiticity and semidefiniteness per block — the trace, the marginal equality in trusted-source mode, and one leg per constraint entry, all decided exactly, on quantities formed in exact arithmetic (Theorem 5.5 (ii)). It is taken on a candidate satisfying D(i)–D(iii) and D(vi), which V1–V3 and V6 decide; a candidate failing one of them is refused there. V7 is the one check of the list that is *not* equivalent to its target condition: it implies D(vii) and is implied by nothing weaker than a candidate that verifies (Theorem 6.2 (iii)).

**V8 (party computability)**
$a_A$ and $a_B$ are tabulated on $\mathcal{O}_\mathsf{A}$, $\mathcal{O}_\mathsf{B}$; and for all $(x, v)$, $(x', v')$ with $\pi_\mathsf{h}(x, v) = \pi_\mathsf{h}(x', v')$ and $a(x, v) = a(x', v')$: $r(x, v) = r(x', v')$.

*The general format.* On a candidate declaration in the general format of Definition 1.1 (DX1)–(DX6) the eight checks read as follows, every operator condition again being decided exactly. In the basic format — one Alice sector, one measured factor, POVM form, two messages neither reading the other and no party-local entry — each reading is the check of the same name above, conjunct for conjunct, so on such a candidate the two readings admit and refuse alike.

*V1 in the general format.* Finiteness and totality as above, every table of the transcript included: each $a_m$ is tabulated on all of $\mathcal{O}_{\pi_m} \times \mathcal{C}_1 \times \cdots \times \mathcal{C}_{m-1}$, which is also where the signature clause of D(viii) lives: a table whose stated domain names a later message's alphabet is not of that type and is refused here. *The sector map is one factorised map*: at most one factor tagged $A$ and, where there is one, first in the list; at least one factor tagged $Q$; *one block kind across the factors*, a disagreement being refused, naming the first factor at which the kind differs; $d_{Q_f} = \sum_{s_f} d_{f,s_f}$ for every factor and $d_Q = \prod_f d_{Q_f}$; and, where an $A$ factor is declared, $d_A = \sum_t d_{A_t}$. The restriction of $\mathsf{idx}$ to each factor's sector set is the total order that factor's own map fixes, and its restriction to $\mathbb{T}$ is the order the $A$ factor's map fixes; the product orders on $\mathbb{S}$ and $\mathcal{V}$ and the order on the joint sectors are derived by (DX1) and (DX3) and are not compared, there being no second datum to compare them with. *And the maps agree, factor by factor*: the map carried by each declared family $\{F^f_{v_f}\}_{v_f}$ is compared against the map of factor $f$, and the map carried by each entry's operator $\hat{O}_k$ against the map of that entry's scope, in the same four components, with the same

refusal naming the first differing position. *Alice's declared data are compared too, against the map of the factor tagged $A$*: the map carried by $\tau_A$, the map carried by each operator of Alice's measurement slot — $M_x$ in POVM form, $N_x$ in ensemble form — and the map carried by each $\Theta_k$ of an entry declared general are each compared against that factor's map in the same four components (block kind, block identifiers, their order, dimensions), with the same refusal naming the first differing position. That comparison is what makes condition (T) of Definition 1.3 decided for Alice's data, which its general-format paragraph calls non-empty for them; it is decided here and nowhere else, and without it a candidate whose Alice block list carried a different block kind, a different identifier list or a different order would pass every check and violate D(i). Where no factor is tagged $A$ there is one Alice block and no map to compare, as in V1 above. *The list of compared objects is again exhaustive by construction*: the objects D(i) quantifies over are the per-factor families, the entries' operators and Alice's declared data. The product family is no datum, and the lifted operator of a scoped entry is no datum; the descriptors that the last links of a chain produce, decided per factor at Definition 7.1 (C1), and a supplied witness, compared at branch (c) of V7, stay outside the list; and the physical families stay outside it too, one per factor.

*V2 in the general format.* The slot holds $\mathsf{none}$, or a block list $(\tau_A^{(t)})_t$ against the $A$ factor's map with $\tau_A^{(t)} = \tau_A^{(t)\dagger}$ and $\tau_A^{(t)} \succeq 0$ for every $t$ and $\sum_t \operatorname{tr} \tau_A^{(t)} = 1$, each trace accumulated over the row index and the block traces then added in the declared order of $\mathbb{T}$, with *every* one of those additions verified to round nowhere; a failure of the last is refused. *Where Alice's measurement slot is in ensemble form, additionally $\tau_A^{(t)} \neq 0$ for every $t$*, a comparison of declared entries with the datum 0, a failure being refused naming the sector. At $|\mathbb{T}| = 1$ this is V2 above, the one block being the whole of $A$.

*V3 in the general format. Per measured factor*: for every $f \in \mathbb{F}$, every $v_f \in \mathcal{V}^f$ and every $s_f \in \mathbb{S}^f$, $F_{v_f}^{f,(s_f)}$ Hermitian and positive semidefinite, and for each $s_f$ the sum $\sum_{v_f} F_{v_f}^{f,(s_f)} = \mathbb{1}_{Q_{f,s_f}}$ with its declared-order sum verified to round nowhere, a failure being refused with the *factor*, the sector and the outcome index at which the first inexact addition occurred. *On Alice's data, block by block along* $\mathbb{T}$: in POVM form $M_x^{(t)}$ Hermitian and positive semidefinite for every $(x,t)$ and $\sum_x M_x^{(t)} = \mathbb{1}_{A_t}$ for every $t$, the sum in the emitted index order verified to round nowhere, a failure being refused with the Alice sector; in ensemble form $N_x^{(t)}$ Hermitian and positive semidefinite for every $(x,t)$ and, *in place of the completeness identity*, the identity $\sum_x N_x^{(t)} = \tau_A^{(t)}$ of (1.6) decided entry by entry against the declared block of $\tau_A$, a failure being refused with the sector and the first differing entry, with that declared-order sum verified to round nowhere, a failure

being refused with the sector and the index of the first inexact addition. *An ensemble-form slot beside a marginal slot holding* none *is refused*; this is D(iii)'s general-format clause, and it is a decision on which of two slots is filled, not on a number. And $\hat{O}_k^{(s_{f_k})}$ is Hermitian for every entry and every block of its scope. *Nothing is decided on a product block* (1.5)*, on a lifted operator or on the denoted family* (1.7): none of the three is a datum, and that the first is a POVM on $Q$ and the third a POVM on $A$ are theorems (Definition 1.1 (DX3), Theorem 5.1 (v)).

*V4, V5, V7 in the general format.* V4 and V5 keep their wording, with $a$ the function (1.8) derives from the transcript's tables; V5's pairs are compared on the values of that function and on nothing about how it was built. V7 keeps its three branches and its refusal, and what its branches (b) and (c) verify is the per-*joint*-sector verification of Theorem 5.5 (ii) in the general format, on a candidate satisfying D(i)–D(iii) and D(vi), in their general-format reading, which V1–V3 and V6 decide (a candidate failing one of them is refused there): the default candidate is (5.2) read per joint sector, and a supplied $\omega^\star$ is required to be a block list over the joint sectors, its map compared in the same four components at branch (c) and there alone.

*V6 in the general format.* When the marginal slot is filled: $\bowtie_k \in \{\preceq, =, \succeq\}$, and the entry's right-hand side is of one of the *three* admitted forms, which with the arriving tag field is what fixes its tag (Definition 2.5): homogeneous with a real scalar, general with a block list $(\Theta_k^{(t)})_t$ along $\mathbb{T}$ decided Hermitian block by block, or party-local with a real scalar. When the slot holds none: the right-hand side is of real scalar type, either untagged or tagged party-local. *The relation symbol is decided by form and not by mode alone*: for an *untagged* entry $\bowtie_k \in \{\preceq, =, \succeq\}$ when the slot is filled and $\bowtie_k \in \{\leq, =, \geq\}$ when it holds none, exactly as in V6 above; for an entry declared party-local, $\bowtie_k \in \{\preceq, =, \succeq\}$ *in either case*, its reading being an operator relation (Definition 2.5, the relation-symbol clause of the party-local form; Definition 1.5, D(vi) in the general format), with the scalar symbols admitted as well where every row of that entry is $1 \times 1$ — every Alice block, and every sector block of every measured factor other than the entry's scope, of dimension one, $d_A = 1$ at one measured factor included — the three operator symbols being the three scalar symbols there. This is a decision on a declared symbol beside a declared tag and measures nothing, and at an entry not declared party-local it is the clause of V6 above word for word, so nothing that clause admits or refuses moves. *Every entry's scope names a measured factor of the emitted sector map*, a comparison of a label against the factor list, a failure being refused naming the entry; this is the general-format half of D(vi) and is decided and never measured, as the relation symbol is. *A party-local entry is the one that arrives carrying that tag value*, and it carries a tag field the type of its datum does not determine, that type not separating it from a homogeneous one. On a candidate carrying

one, the arriving-tag clause of V6 above therefore has a subject, and it is this comparison: the datum is of the type *that* tag admits, a real scalar, and the scope names a measured factor; a party-local tag beside a matrix right-hand side is refused. The sentence of V6 above that an entry holds no tag field independent of the type of its datum is true of the two untagged forms, homogeneous and general, and of them alone. *Nothing is inferred*: an entry arriving without the party-local tag is never read party-locally. So the arriving-tag clause refuses no party-local entry beside a real scalar, which a literal reading of it would otherwise do, and wherever no entry is tagged the reading is that of V6 above (Definition 2.5).

*V8 in the general format.* The announcement clause reads the transcript: every $a_m$ is tabulated on all of its stated domain, which is V1's clause restated where D(viii) pairs with it; the key clause is unchanged, with $a$ the derived function.

**Theorem 6.2 (Soundness, completeness and decidability of the validator).** For a candidate declaration with finite tabulated data:

(i) *(soundness, on all eight conditions)* if V1–V8 all pass then D(i)–D(viii) all hold, of the declared numbers;

(ii) *(completeness, on seven of the eight)* if D(i)–D(vi) and D(viii) hold then V1–V6 and V8 all pass;

(iii) *(V7 is one-sided, and the gap is exactly its third branch)* the converse of (i) fails, and it fails only at V7: D(vii) does *not* imply V7. For a witness, take untrusted-source mode with $Q$ a qubit carrying the single sector $\mathbb{S} = \{s_0\}$, every other slot well-formed, and the one-entry list $\mathfrak{L} = \{(|0\rangle\langle 0|_Q\,,\ \geq,\ 1)\}$, whose entry reads $\mathrm{tr}[(\mathbb{1}_A \otimes |0\rangle\langle 0|)\,\omega] \geq 1$. Then $\hat{O}_1$ is a one-block list, Hermitian, the typing is that of D(vi), and $\omega = \eta \otimes |0\rangle\langle 0|$ is feasible for any state $\eta \in S_=(A)$ — it is Hermitian and positive semidefinite outright, has unit trace, and its constraint value is exactly 1 — so $\mathcal{W} \neq \varnothing$ and D(i)–D(viii) all hold. But $\mathfrak{L} \neq \varnothing$, and the default candidate (5.2) that branch (b) of V7 tries first gives $\mathrm{tr}[(\mathbb{1}_A \otimes |0\rangle\langle 0|)(\mathbb{1}_A/d_A \otimes \mathbb{1}_Q/2)] = \frac{1}{2}$, which fails $\geq 1$; so with no witness supplied branch (c) is unavailable, V7 fails and the well-formed declaration is refused. Exactly the same happens for every declaration with $\mathfrak{L} \neq \varnothing$ presented without a witness whose feasible set is non-empty but does not contain (5.2). The refusal errs on the safe side — a refused declaration is never consumed — and the incompleteness is deliberate: deciding D(vii) in general is a semidefinite feasibility problem this module does not solve;

(iv) *(decidability, and what is computed)* each of V1–V6 and V8 is decided by finite enumeration over the declared alphabets together with finitely many arithmetic operations on the declared matrices; V7 is decided in

the same sense on the candidate of its branch (b) or (c), and is otherwise a refusal. The arithmetic part consists of *exact decisions*: an identity, a semidefiniteness, or the exactness of an addition. The following counts give the complete applicable predicate inventory for well-typed data, not the number an early-terminated execution necessarily performs. Write $I_{\mathrm{tr}} = 1$ in trusted-source mode and $I_{\mathrm{tr}} = 0$ when the marginal slot holds none. The complete arithmetic inventory of V2, V3 and V6 is

$$\underbrace{3I_{\mathrm{tr}}}_{\text{V2}} + \underbrace{2|\mathcal{X}| + 1}_{\text{V3 on } A} + \underbrace{|\mathbb{S}|\,\big(2|\mathcal{V}| + 1\big)}_{\text{V3 on } Q} + \underbrace{m\,|\mathbb{S}|}_{\text{V3 on } \mathfrak{L}} + \underbrace{m_{\mathrm{gen}}}_{\text{V6}} \tag{6.1}$$

predicate decisions, where $m_{\mathrm{gen}} \le m$ counts the entries declared general — the only ones carrying a matrix whose Hermiticity is decided; V6's remaining content is $m$ comparisons of a relation symbol, a tag and a type, which are decided and not computed. *Beside these stand the summation decisions of condition (S)*: one on $\operatorname{tr} \tau_A$ in trusted mode and none on the absent marginal in untrusted mode; one on $\sum_x M_x$; and one per sector on $\sum_v F_v^{(s)}$, so $|\mathbb{S}| + 1 + I_{\mathrm{tr}}$ of them, each a pass over the additions of one declared-order sum. And V7, on the candidate it verifies, takes $2|\mathbb{S}| + 1$ decisions — Hermiticity and semidefiniteness per block, and the trace — one further in trusted mode for the marginal equality, and one decision per constraint entry: semidefiniteness for an inequality, exact entrywise equality for the relation $=$ (Theorem 5.5 (ii)); in untrusted mode a constraint leg is one comparison of real numbers instead. Every one of them is taken on a matrix of size $d_A$, $d_s$ or $d_A d_s$ for a single $s$; *the validator assembles no array larger than one such matrix*, which where $|\mathbb{S}| = 1$ is $d_Q$ or $d_A d_Q$ square, the one block being the whole space there (Theorem 8.2 (i)). In trusted mode the V2 term is 3 and not 4: D(ii)'s named Hermiticity condition and the Hermiticity half of its positivity condition are the same decision, taken once. In untrusted mode V2 decides only the marginal slot's none branch, with no marginal arithmetic or marginal summation predicate.

*The general format.* All four parts hold for a candidate declaration in the general format of Definition 1.1 (DX1)–(DX6), with the checks read as in the general-format paragraph of Definition 6.1 and the conditions as in that of Definition 1.5: (i) soundness on all eight and (ii) completeness on D(i)–D(vi) and D(viii); (iii) with the *same* witness, which is a general-format declaration at $|\mathbb{T}| = 1$, one measured factor and no party-local entry; and (iv) with the predicate decisions of V2, V3 and V6 numbering

$$\underbrace{I_{\mathrm{tr}}\big(2|\mathbb{T}| + 1 + |\mathbb{T}|\,\nVdash[\text{ensemble form}]\big)}_{\text{V2}} + \underbrace{|\mathbb{T}|\,\big(2|\mathcal{X}| + 1\big)}_{\text{V3 on } A} + \underbrace{\sum_{f \in \mathbb{F}} |\mathbb{S}^f|\big(2|\mathcal{V}^f| + 1\big)}_{\text{V3 per measured factor}} + \underbrace{\sum_{k=1}^{m} |\mathbb{S}^{f_k}|}_{\text{V3 on } \mathfrak{L}} + \underbrace{m_{\mathrm{gen}}\,|\mathbb{T}|}_{\text{V6}}, \tag{6.2}$$

beside them the summation decisions of condition (S) — one on $\operatorname{tr}\tau_A$ in trusted mode only, one per Alice sector on Alice's family, and one per factor and sector on that factor's family, so $|\mathbb{T}| + \sum_f |\mathbb{S}^f| + I_{\mathrm{tr}}$ of them — and V7, on the candidate it verifies, $2|\mathbb{T}|\,|\mathbb{S}|+1$ decisions, $|\mathbb{T}|$ further in trusted mode for the marginal equalities, and per entry $k$ either $|\mathbb{T}|$ operator comparisons, one comparison of real numbers in untrusted mode, or, for a party-local entry, $|\mathbb{T}|\,|\mathbb{S}|/|\mathbb{S}^{f_k}|$ operator comparisons. Each operator comparison is one semidefiniteness decision for an inequality or one exact entrywise equality decision for $=$ (Theorem 5.5 (ii), general format). *Every one of them is taken on a matrix of size $d_{A_t}$, $d_s$ or $d_{A_t}d_s$ for a single joint sector, or of size $d_{A_t}d_s/d_{f,s_f}$ for a single party-local row; the validator assembles no array larger than one joint block.* In the basic format — $|\mathbb{T}| = 1$, one measured factor, POVM form, two messages neither reading the other and no party-local entry — (6.2) is (6.1) term for term, and every other count of this paragraph is the corresponding count of (iv).

# 7 Consuming the reduction chain

**Definition 7.1** (Interface conditions on a chain)**.** A reduction chain $\mathfrak{R} = (\mathfrak{R}_{\mathrm{src}}, \mathfrak{R}_{\mathrm{rec}})$ presented to the compiler, with $\mathfrak{R}_{\mathrm{src}} = (\Xi_1, \ldots, \Xi_{L_{\mathrm{s}}})$ and $\mathfrak{R}_{\mathrm{rec}} = (\Lambda_1, \ldots, \Lambda_L)$, satisfies the *interface conditions on a chain* when the following hold.

*Receiver segment.*

(C1) *(descriptor match)* the target descriptor of $\Lambda_i$ equals the source descriptor of $\Lambda_{i+1}$ for $1 \le i < L$; the source descriptor of $\Lambda_1$ is the physical receiver space of $\mathfrak{S}_{\mathrm{rec}}$; and the target descriptor of $\Lambda_L$ is $(Q, \mathfrak{s})$ — that is, the last link's target descriptor *is* the sector map. That equality is decided by the same four-component comparison V1 makes on the map of a declared $Q$-operator family (Definition 6.1);

(C2) *(outcome correspondence)* the declared outcome correspondence is a tabulated total surjection $\gamma : \mathcal{V}^{\mathrm{ph}} \to \mathcal{V}$ from the *physical* alphabet onto the *emitted* one — the coarse-graining direction, of which a bijection is the special case;

(C3) *(target POVM)* $\{F_v\}_{v\in\mathcal{V}}$ is a family of block lists against $\mathfrak{s}$ — *typing*, held by the emitted form and not checked — and for each $s$ the block family $\{F_v^{(s)}\}_v$ is a POVM on $Q_s$, decided *exactly* on the emitted blocks themselves: $F_v^{(s)}$ Hermitian and positive semidefinite for each $v$, $\sum_v F_v^{(s)} = \mathbb{1}_{Q_s}$, and the declared-order sum forming it rounding nowhere (Definition 1.3). This condition is check V3 restricted to $\{F_v\}$ and states the same list of conjuncts (Theorem 7.5 (ii));

(C4) *(pullback, where checkable)* if every link of $\mathfrak{R}_{\mathrm{rec}}$ supplies its channel as data then, with $\Lambda := \Lambda_L \circ \cdots \circ \Lambda_1$,

$$\Lambda^{\dagger}(F_v) = \sum_{w \in \gamma^{-1}(v)} F_w^{\mathrm{ph}} \qquad \text{for every } v \in \mathcal{V}, \tag{7.1}$$

an *exact* matrix identity, decided entry by entry on the emitted matrices and on the supplied channel data, with the fibre sum accumulated in the declared order of $\mathcal{V}^{\mathrm{ph}}$. Where some link supplies no channel as data, only a theorem that constructs one, the identity is not decidable from the emitted data and is carried instead as an obligation of the chain's producer, Assumption 10.6;

(C5) *(slots)* each link carries its constraint entries and its licence slot, both possibly empty, and its status tag.

*Source segment.* The source segment satisfies the interface conditions (S1)–(S4) of Definition "Source segment and its interface conditions" (transmitter/def:source-segment) — descriptor match with the virtual transmitted system $A''$ as the source descriptor of $\Xi_1$, on which the certificate of Definition "Source-replacement certificate" (transmitter/def:srcert) is written, and every descriptor Alice-side; index preservation; the source identity $\Xi(\rho_x^{\mathrm{v}}) = \rho_x$ for every $x$, that definition's equation (S3), where every link supplies its channel as data; and the slots — *decided by this module on the presented data*, exactly as that definition states them. When $L_{\mathrm{s}} = 0$ the segment is empty, $\Xi$ is the identity, $A'' = A'$ and $\rho_x^{\mathrm{v}} = \rho_x$, so (S3) holds by inspection.

*Between the segments.*

(X1) *(order)* the chain is composed source segment first and receiver segment second. This is not a tag and is not checked: it is the shape of the record. A source-map link has no slot in $\mathfrak{R}_{\mathrm{rec}}$ to be written into, and a link presented in the wrong segment fails that segment's descriptor rule — (C1) for a source map, since $A'$ is not the physical receiver space, and (S1) for a squashing map, since $Q$ is not an Alice-side descriptor.

*At which reading (C4) and (S3) are decided.* Both are *matrix equalities on declared data* — a composite channel supplied as binary64 data, applied to binary64 POVM elements on one side and to binary64 signal states on the other — and both are decided *exactly*, entry by entry, on the numbers the compiler holds. *The composite, the adjoint and the fibre sum are formed in exact arithmetic as well*, so the identity decided is the identity written; that the operands are binary64 is not what makes the decision exact, a product of two binary64 numbers need not be one. Exactness here is an obligation of the chain's producer (Assumption 10.6): a producer whose composite lands off the identity by a rounding has not delivered the reduction the interface

asks for, and repairs its own emission; no window is opened here. Everything else in this definition — descriptors, alphabets, fibres, slots, segments — is combinatorial and is read exactly, as D(vi)'s relation symbols are.

*The general format: one receiver segment per measured factor.* For a chain presented as Definition 3.1 (c′), with $\mathfrak{R}_{\mathrm{rec}} = (\mathfrak{R}^f_{\mathrm{rec}})_{f\in\mathbb{F}}$, the conditions (C1)–(C5) are decided *of each segment separately, and of nothing across two segments*. (C1): within segment $f$ the target descriptor of a link is the source descriptor of the next, the source descriptor of $\Lambda^f_1$ is factor $f$'s physical receiver system, and the target descriptor of $\Lambda^f_{L_f}$ is $(Q_f, \mathbb{S}^f, (d_{f,s_f})_{s_f})$ — *that factor's own sector map, and not the product map*, which is derived by Definition 1.1 (DX1) and is the target descriptor of nothing; the same four-component comparison decides it. (C2): one tabulated total surjection $\gamma_f$ from factor $f$'s physical alphabet onto the alphabet factor $f$'s last link delivers. (C3): the target family $\{F^f_{v_f}\}$ is a family of block lists against factor $f$'s map and, for each $s_f$, a POVM on $Q_{f,s_f}$ decided exactly on that factor's blocks — which is check V3 restricted to that factor, conjunct for conjunct, in its general-format reading. (C4): where every link of segment $f$ supplies its channel as data, with $\Lambda_f \coloneqq \Lambda^f_{L_f} \circ \cdots \circ \Lambda^f_1$,

$$\Lambda_f^{\dagger}\big(F^f_{v_f}\big) = \sum_{w_f \in \gamma_f^{-1}(v_f)} F^{\mathrm{ph},f}_{w_f} \qquad \text{for every } v_f, \tag{7.2}$$

an exact matrix identity on that factor's matrices, the fibre sum accumulated in the declared order of that factor's physical alphabet. *The product composite* $\bigotimes_f \Lambda_f$ *and the product identity on* $\{F_v\}$ *are formed and decided by no one*: the identity $\big(\bigotimes_f \Lambda_f\big)^{\dagger}(F_v) = \bigotimes_f \Lambda_f^{\dagger}(F^f_{v_f})$ is a consequence of (7.2) and of the product rule (1.5) — it is Theorem "The emission of several measuring parties" (reductions/thm:emitted-product) (ii), named here and decided here by nothing — and (C4) is the per-factor identity and nothing wider. (C5) is per link, as above. Where some link of segment $f$ supplies no channel as data, only a theorem that constructs one, (7.2) for *that* segment is not decidable from the emitted data and is carried as Assumption 10.6, exactly as above and per factor.

*A non-empty source segment, and the conditions it brings. A presented chain may carry one.* The rule by which a non-empty source segment composes with a receiver segment is a theorem of the reductions module (Lemma "Licensed admissibility survives precomposition with a source channel" (reductions/lem:precomposition), Theorem "The two-sided reduction in trusted-source mode" (reductions/thm:two-sided), with Definition "Reduction chain" (reductions/def:chain)), in trusted-source mode; and the module that admits source-link kinds may emit an artefact carrying a link of an admitted kind. What follows is what this module decides of such a segment, on the presented data and exactly; whether a presented chain carries one is data of the record and is asserted here of no artefact.

- *(S1)–(S4) as Definition "Source segment and its interface conditions" (transmitter/def:source-segment) states them*, which is what the source-segment paragraph above already says; and within (S3), *the split by kind of witness*: for a link supplying its channel as data the per-link identity is decided here, entry by entry, with the composite formed in exact arithmetic; for a link supplying a theorem constructor it is *not decided here at all* but is the conclusion of the claim that link's obligation record names, and the composite identity is the composition of the per-link ones. So what this module decides of (S3) is the data links' share of it, and the constructor links' share travels as the obligation Assumption 10.6 names — Theorem 7.5 (iv), (v) — and is discharged nowhere in this contract.

- *(A1)–(A3) of that definition, which are decided on data and are not part of (S1)–(S4)*: that the triple a link's obligation record carries — its kind, the claim id it names, and the assumption short ids it names — appears *entry by entry* in the soundness registry the artefact carrying the segment emits, both components of that registry entry's assumption list compared, so that a record naming a claim's closure and omitting the source model's attestation fails as one naming neither does; that every assumption of that entry is in the artefact's assumption ledger; and that the segment is presented inside an artefact of the module that admits its links. These are comparisons of finitely many declared strings and of a finite list against a finite list, and *they are decided here because they are decidable on the presented record*, which is why they are named beside (S1)–(S4) rather than left to Assumption 10.6. *What they do not decide* is that the named claim's conclusion applies to these descriptors — family membership is the emitting module's assertion and no consumer re-proves a theorem — and a link supplying its channel as data carries no obligation record and is decided by (S3) instead.

- *the licensed-only rule.* Beside a non-empty source segment *every* entry of the emitted $\mathfrak{L}$ is required to carry a real scalar as its right-hand side — homogeneous, party-local, or an untrusted-mode scalar (Definition 2.5).

- *the declared source mode. A declaration carrying a non-empty source segment is in trusted-source mode*: the rule fails where the marginal slot of (1.1) holds none (Definition 1.1), the untrusted-source mode, while $L_{\mathrm{s}} \geq 1$.

*(X1) in the general format.* The order rule is unchanged and is still the type of the record: a source-map link has no slot in any receiver segment to be written into, and a link presented in the wrong list fails that list's

descriptor rule — (C1) of the factor whose segment it was written into, or (S1). In the reductions module the composite of a chain is a pair, and the reason a source link may not follow a receiver link is a theorem of that module (Lemma "Licensed admissibility survives precomposition with a source channel" (reductions/lem:precomposition), Theorem "The two-sided reduction in trusted-source mode" (reductions/thm:two-sided)) beside the published warrant [1, Rem. 9.2]; this module types the record, decides the descriptors, and states neither.

**Theorem 7.5** (Decided identities and named obligations of a consumed chain)**.** Let a chain be presented to the compiler, with the interface conditions of Definition 7.1. Then:

(i) (C1)–(C5) and (S1)–(S4) are decided by finite enumeration and finitely many arithmetic operations on the emitted data, and a chain failing any of them is refused. (X1) is not decided, because it is not a condition on data: it is the type of the record;

(ii) (C3) is exactly the part of D(iii) concerning $\{F_v\}$, in both halves — typing against $\mathfrak{s}$, and the per-block predicates and summation condition of Definition 1.3 — hence a chain passing the interface conditions yields an emitted $\{F_v\}$ for which Theorem 3.7 applies. Where the Bob-side weighting of Definition 4.1 is applied after the chain (Definition 3.3), (C3) is decided on the family the chain delivered, the emitted family is its weighting $N_{v,f} = P(f \mid t(v))F_v$, check V3 decides the emitted family itself (Theorem 3.7 (i)), and Lemma 4.2 (i), (ii) is what says the weighting cannot break what (C3) established; the identity of conjunct lists asserted here is between (C3) and D(iii) on the chain's family;

(iii) *(what a verified (C4) buys)* when the composite receiver channel $\Lambda$ is given as data, (C4) is *equivalent* to the statement that measuring the emitted family $\{F_v\}$ after $\Lambda$ reproduces the $\gamma$-*coarse-grained* physical measurement statistics:

$$\operatorname{tr}[F_v\,\Lambda(\rho)] = \sum_{w\in\gamma^{-1}(v)} \operatorname{tr}\left[F^{\mathrm{ph}}_w\,\rho\right] \tag{7.3}$$

for every $v \in \mathcal{V}$ and every operator $\rho$ on the physical receiver space. Consistency with completeness holds *under one hypothesis this contract names and does not verify*: if the supplied channel data are trace-preserving, so that $\Lambda^\dagger$ is unital, then summing (7.1) over $v$ returns completeness of the physical family and nothing more, whereas the same identity read with a coarse-graining in the opposite direction — a total surjection $\gamma' : \mathcal{V} \to \mathcal{V}^{\mathrm{ph}}$, with $\Lambda^\dagger(F_v) = F^{\mathrm{ph}}_{\gamma'(v)}$ — does not in general, forcing every physical element covered more than once by $\gamma'$

to vanish. That trace preservation is Assumption 10.6, not a check: nothing here evaluates it;

(iv) *(what a verified (S3) buys)* when the composite source channel $\Xi$ is given as data, (S3) of Definition "Source segment and its interface conditions" (transmitter/def:source-segment) — the source identity, that definition's equation, decided by this module on the presented data (Definition 7.1) — says that the virtual source, pushed through $\Xi$, is the specified source: $\Xi(\rho_x^{\mathrm{v}}) = \rho_x$ for every $x$, hence

$$(\mathrm{id}_X \otimes \Xi)\Big(\sum_{x\in\mathcal{X}} p_x\,|x\rangle\langle x|_X \otimes \rho_x^{\mathrm{v}}\Big) = \sum_{x\in\mathcal{X}} p_x\,|x\rangle\langle x|_X \otimes \rho_x, \tag{7.4}$$

and, averaging, $\sum_x p_x \Xi(\rho_x^{\mathrm{v}}) = \sum_x p_x \rho_x$: the prepare-and-measure states of the emitted virtual description map onto the specified ones, index by index, with the weights unchanged by (S2). The compiler decides the source identity (S3) and nothing beyond it; and

(v) the following are *named and never discharged here*: that each link of either segment is a sound reduction — that its channel exists as claimed for a link supplying only a theorem constructor, that a link supplying its channel as data supplies a completely positive and trace-preserving map, and that every reachable state after the reduction satisfies every transported constraint entry — and that any restricted attack set claimed by a link is licensed. These are obligations of the chain's producer, the soundness of each link being Assumption 10.6. No claim of this contract discharges them for a presented chain in general — for the Band A identity link, which Definition 8.1 (b) fixes with its channel supplied as data, its outcome correspondence and its target POVM, Theorem 8.2 (v) discharges every clause of Assumption 10.6 by exhibition — and no proof below invokes them.

*The general format.* For a chain presented as Definition 3.1 (c′), with the interface conditions read as in the general-format paragraph of Definition 7.1, (i)–(v) hold with the following readings, of which the five above are the case of one measured factor and an empty source segment.

(i′) (C1)–(C5) are decided per measured factor and (S1)–(S4) as before, all by finite enumeration and finitely many arithmetic operations, and a chain failing any of them is refused. *And (A1)–(A3), the licensed-only rule and the declared-source-mode rule are decided as well*: the first three by comparing finitely many declared strings and one finite list against another, the fourth by reading $m$ declared tags, and the fifth by reading one declared slot. (X1) is again not decided, being the type of the record.

(ii′) (C3) is exactly the part of D(iii) concerning the declared measured families, *factor by factor*: for each $f$, (C3) read for $f$ and D(iii)'s clauses on $\{F^f_{v_f}\}$ are the same list of conjuncts, so a chain passing the interface conditions yields per-factor families for which Theorem 3.7 (i′) applies. *That the product family is a POVM on $Q$ is on neither side of that pairing*, being a consequence of the factors' conditions and no condition on data (Definition 1.1 (DX3), Theorem 6.2, general format). The Bob-side weighting sentence applies per factor, to that factor's delivered family.

(iii′) *(what a verified (C4) buys, per factor and jointly)* when every link of segment $f$ supplies its channel as data, (7.2) is equivalent to $\mathrm{tr}\left[F^f_{v_f}\Lambda_f(\rho_f)\right] = \sum_{w_f\in\gamma_f^{-1}(v_f)}\mathrm{tr}\left[F^{\mathrm{ph},f}_{w_f}\rho_f\right]$ for every $v_f$ and every operator $\rho_f$ on factor $f$'s physical receiver space. *And the joint statement follows for every operator, product or not*: if every segment supplies its channels as data then, with $\Lambda \coloneqq \bigotimes_f \Lambda_f$, $\gamma \coloneqq \prod_f \gamma_f$ and $F^{\mathrm{ph}}_w \coloneqq \bigotimes_f F^{\mathrm{ph},f}_{w_f}$,

$$\Lambda^\dagger(F_v) = \sum_{w\in\gamma^{-1}(v)} F^{\mathrm{ph}}_w, \qquad \text{hence} \qquad \mathrm{tr}[F_v\,\Lambda(\rho)] = \sum_{w\in\gamma^{-1}(v)} \mathrm{tr}\left[F^{\mathrm{ph}}_w\,\rho\right] \tag{7.5}$$

for every $v\in\mathcal{V}$ and every operator $\rho$ on the product physical space. *Neither side of (7.5) is formed or decided by anything*: it is a consequence of the per-factor identities, and the objects in it — the product channel, the product physical family, the product target family — are data of no one. The consistency reading holds under the same hypothesis read per factor: if every factor's supplied channel data are trace preserving, so that each $\Lambda_f^\dagger$ is unital, then so is $\Lambda^\dagger$ and summing (7.5) over $v$ returns completeness of the product physical family and nothing more. That trace preservation is Assumption 10.6, per factor, and is evaluated nowhere.

(iv′) (S3) is as in (iv), with the split by kind of witness: what a *decided* (S3) buys is (7.4) and its averaged form, for the data links' share of the composite; for a link supplying a theorem constructor this module decides nothing and what its share buys is what the claim its obligation record names concludes, which this contract neither states nor asserts.

(v′) the list of obligations named and never discharged here gains three items, each per the general format: that each link of *each* receiver segment is a sound reduction, factor by factor; that a non-empty source segment composes with the receiver segments as a reduction, which is Theorem "The two-sided reduction in trusted-source mode" (reductions/thm:two-sided), used by no step here; and that the claim an obligation record names applies to that link's descriptors, which

(A1)–(A3) do *not* decide. These are again the obligations of (v), Assumption 10.6 and the attack-set licence, and the emitting module's own assertion; no claim of this contract discharges them, save as (v) says of the Band A identity link, and no proof below invokes them.

# 8 The Band-A route

**Definition 8.1** (Band-A specification)**.** A specification is *Band A* when (a) its receiver part declares a POVM $\{F_w^{\mathrm{ph}}\}_{w\in\mathcal{V}^{\mathrm{ph}}}$ directly on a finite-dimensional system, with no characterization step and no photon-number structure, and the emitted sector set is the singleton $\mathbb{S} = \{s_0\}$ with $\Pi_{s_0} = \mathbb{1}_Q$; and (b) its reduction chain has an *empty source segment*, $L_{\mathrm{s}} = 0$ — the source segment of Definition "Source segment and its interface conditions" (transmitter/def:source-segment) at length zero, required here of a Band-A chain and of no other — and a receiver segment of one link, $L = 1$, $\Lambda_1 = \mathrm{id}_Q$ supplied as data, with status *exact*, empty constraint slot and empty licence slot, whose outcome correspondence is the identity map $\gamma = \mathrm{id}$ on $\mathcal{V} = \mathcal{V}^{\mathrm{ph}}$ and whose target POVM is the declared family itself: $F_v = F_v^{\mathrm{ph}}$, entry for entry, for every $v \in \mathcal{V}$.

**Theorem 8.2** (The Band-A route)**.** For a Band-A specification:

(i) the block structure of a Band-A declaration is the one-block structure: $\mathfrak{s}$ is the singleton map $\big((s_0, d_{s_0})\big)$, $d_Q = d_{s_0}$, and every $Q$-operator is a one-element block list, so that its single block *is* the whole operator. Two consequences follow, and they differ. The typing half of D(iii) is not vacuous but trivial to satisfy: an operator is a one-element list, and V1's sector-map comparison is one identifier and one dimension. The five predicates of D(iii) are *not* vacuous at all — with $|\mathbb{S}| = 1$ the per-block predicates are the full-space ones, decided once each on $d_Q \times d_Q$ matrices, and a Band-A declaration is decided exactly like any other, condition (S) included;

(ii) the interface conditions of Definition 7.1 hold by inspection, save the predicates and summation condition of (C3), which are decided: descriptors match because source and target of id coincide; the outcome correspondence is the identity bijection $\gamma = \mathrm{id}$ on $\mathcal{V} = \mathcal{V}^{\mathrm{ph}}$, as Definition 8.1 (b) requires; (C3)'s typing half holds because the declared receiver family is already a one-element block list against the singleton map, and its predicates and summation condition are the declared receiver POVM's own, so (C3) holds exactly when that family is exactly admissible, which is decided on it and is not given by inspection; the pullback identity (C4) reads $\mathrm{id}^\dagger(F_v) = F_v = F_v^{\mathrm{ph}}$ and holds *identically* — entry for entry, since $\mathrm{id}^\dagger = \mathrm{id}$, every fibre $\gamma^{-1}(v)$ is a singleton,

and the delivered $F_v$ *is* the declared $F_v^{\mathrm{ph}}$, the same matrix, by Definition 8.1 (b); the source conditions (S1)–(S4) of Definition "Source segment and its interface conditions" (transmitter/def:source-segment) are vacuous, quantifying over an empty list of links — the empty source segment being that definition's at $L_{\mathrm{s}} = 0$, which Definition 8.1 (b) requires of a Band-A chain and which is therefore a hypothesis of this theorem — and the virtual ensemble of (S3) is the specified one, so the source identity reads $\rho_x = \rho_x$; the ordering (X1) holds because the source segment is empty; and the slots of (C5) are present and empty. The one link of the receiver segment is the identity link. *So both conditions that a general chain has to decide on computed data are met here by exhibition*, and what a verified pullback identity and a verified source identity buy on a general chain holds on this path without appeal to any obligation of the chain's producer: measuring the family the link delivers after id reproduces the physical statistics, $\mathrm{tr}[F_v\,\mathrm{id}(\rho)] = \mathrm{tr}\left[F_v^{\mathrm{ph}}\,\rho\right]$ for every $v$ and every operator $\rho$, and id is exactly trace-preserving *by exhibition*, the supplied channel data being exactly that map rather than binary64 data trusted to be trace-preserving, so its adjoint is unital and summing the pullback identity over $v$ returns $\sum_v F_v^{\mathrm{ph}} = \mathbb{1}$, which on this path is the same statement as (C3)'s own completeness, $\{F_v^{\mathrm{ph}}\}$ being $\{F_v\}$: the identity holds as a triviality and carries no content this path did not already have;

(iii) the link transports no constraint entry, so $\mathfrak{L}$ is the specification's own list $\mathfrak{S}_{\mathrm{con}}$ and the constraint-soundness obligation of the link (the transported-constraint clause of Assumption 10.6) is vacuous; where $\mathfrak{S}_{\mathrm{con}}$ is empty, $\mathfrak{L} = \varnothing$ and the two-part gate of Theorem 9.4 is not engaged;

(iv) consequently no emitted slot, and no claim of this contract used in the Band-A path, depends on an artefact of the receiver module or of the reductions module. The residual physical content of the receiver is Assumption 10.8, and it is an idealization of the receiver, not a reduction of it; an entry the specification declares itself engages the gate, and with it the attack-set licence of Assumption 10.7, as on every route; and

(v) *(the identity link is sound)* the Band-A chain satisfies every condition of Assumption 10.6: (i)(a) the channel of its one link, $\mathrm{id}_Q$, exists, the link supplying it as data; (i)(b) that link satisfies the pullback identity (7.1); (i)(c) the protocol using the measurement the link delivers is the protocol using the declared receiver family $\{F_v^{\mathrm{ph}}\}$, the delivered $\{F_v\}$ being that family, so security of the one is security of the other, and where a setting-dependent weighting $P(f\,|\,t)$ is applied after the link (Definition 4.1), the emitted family $\{P(f\,|\,t(v))\,F_v\}$ is the identi-

cally weighted declared family $\{P(f \mid t(v))\, F_v^{\mathrm{ph}}\}$, so the same holds of the weighted protocol; (ii) its source segment is empty; (iii) its link transports no constraint entry; and (iv) $\mathrm{id}_Q$ is completely positive and trace-preserving. No clause of Assumption 10.6 is therefore a premise on this path; that the declared family is the measurement the receiver performs is Assumption 10.8, and no condition on the link.

# 9 The emission gates

**Definition 9.1** (Key-length eligibility)**.** An emitted declaration is *key-length-eligible* when all of the following hold:

(G1) V1–V8 all pass — whereupon D(i)–D(viii) all hold of the declared numbers, by the soundness half of Theorem 6.2 (i);

(G2) *(a consumption gate on the transmitter artefact)* if the marginal slot is filled and $d_A \geq 2$: the transmitter artefact carries the source-replacement licence record of Definition "Source-replacement certificate" (transmitter/def:srcert) — the certificate $(\tau_A, \Psi, \{p_x, \rho_x^{\mathrm{v}}\})$ with its attestation (SR3) — whose *five decidable conditions* the transmitter decided exactly on the same data (Theorem "Checkability and sufficiency of the source-replacement certificate" (transmitter/thm:certificate-sufficiency) (i)) and which *this module re-decides on the emitted data*: $\Psi = \Psi^\dagger$; $\Psi^2 = \Psi$; $\mathrm{tr}\,\Psi = 1$, the trace accumulated over the row index and rounding nowhere; (SR1); and (SR2) — each *exactly*, as matrix and scalar identities. The transmitter's decision is not taken on its word. *No condition of (G2) is stated on a vector.* The certificate's declared datum is the operator $\Psi$; the unit vector $|\psi\rangle$ with $\Psi = |\psi\rangle\langle\psi|$ exists by $\Psi^2 = \Psi$ and $\mathrm{tr}\,\Psi = 1$ and is never itself declared (Definition "Source-replacement certificate" (transmitter/def:srcert)). Clause (SR3) carries no number and is not decidable (Theorem 9.4 (i)): (G2) asks for its presence only. At $d_A = 1$ no record is demanded, the licence being vacuous there;

(G3) if $m \geq 1$: (a) $\mathfrak{L}$ lies in the licensed class of the emitted source mode (Definition 2.5) — *a type check on each entry's right-hand-side tag, comparing no numbers — and* (b) the attack-set licence record, the attestation of Assumption 10.7, is present.

*(G2) by form of the certificate.* The five conditions above are those of a certificate declared in pure form: a pure-form certificate is re-decided exactly as stated, and refused when one of the five fails.

A certificate declared in block form is refused at this gate by its declared form, whatever its blocks, by row 3 of the gate outcome table of Definition 9.3. A block-form certificate (Definition "Source-replacement certificate" (transmitter/def:srcert)) is a positive semidefinite unit-trace block list on

which that definition imposes no purity, so one with more than one non-zero block, or whose one non-zero block is not a rank-one projector, also fails $\Psi^2 = \Psi$; but the refusal does not rest on that, since one whose one non-zero block is a rank-one projector can meet all five conditions above.

An ensemble presentation declares no $\Psi$ and no $\{M_x\}$ to re-decide. For it the gate decides, in place of the five conditions, what the presentation declares: (E1), (E2) and (E3) of Definition "Source-replacement certificate" (transmitter/def:srcert) on the declared weighted ensemble, and that the emitted declaration is in ensemble form with its Alice slots *the* ones the presentation defines — the sector structure of the factor tagged $A$ is the matched structure, and for every $t$ and $x$ the block $N_x^{(t)}$ is entry for entry the transpose of the declared block $\sigma_x^{\mathrm{v},(t)}$ and $\tau_A^{(t)}$ the transpose of their declared-order sum. *A presentation accompanying a POVM-form declaration is refused*, nothing tying the declared $\{M_x\}$ to it, which is the transmitter's own rule for a POVM-form consumer. In every form, (G2) asks for the presence of (SR3).

*(G3)(a) with the party-local form.* The licensed class of Definition 2.5 contains, in either source mode, the entries declared party-local; the decision remains a type check on each entry's tag and compares no numbers. A list carrying such an entry is key-length-eligible only with the attack-set licence record of (b), as every non-empty list is, and what that record attests for a tagged entry is a certificate of the module that produces the chain, on the scoped factor's own channel (Definition 2.5).

**Definition 9.3** (The gate outcome table)**.** The *gate outcome table* is the following table of the failures of the gates of Definition 9.1 and their outcomes; where the failures of one declaration meet the premises of a refusal row and of an emission row both, the outcome is the refusal.

*Gate failure to emitted outcome.* No row substitutes a default value, drops an entry, changes the source mode or synthesizes an attestation.

| *What failed* | *Outcome* |
|---|---|
| (G1): any of V1–V8 | refusal, naming the check |
| (G2): trusted mode at $d_A \geq 2$, certificate *absent*, and $m = 0$ | none (no outcome of the compiler) |
| (G2): trusted mode *at any* $d_A$, certificate *present* but one of the five conditions of Definition 9.1 (G2) fails — $\Psi$ Hermitian, $\Psi^2 = \Psi$, $\operatorname{tr}\Psi = 1$ rounding nowhere, (SR1), (SR2) | refusal, naming the condition |
| (G2): trusted mode at $d_A \geq 2$, certificate absent, and $m \geq 1$ | none (no outcome of the compiler) |
| (G3)(a): $\mathfrak{L}$ not in the licensed class — some entry declared general in trusted-source mode | emission, not key-length-eligible |
| (G3)(b): the attack-set licence record absent | emission, not key-length-eligible |

The cell of rows 2 and 4 assigns the compiler no outcome.

**Theorem 9.4 (The emission gates and their outcomes).** (i) *(decidability)* (G1) is decidable by Theorem 6.2; (G2)'s conditions — the five of Definition 9.1 (G2), namely $\Psi$ Hermitian, $\Psi^2 = \Psi$, $\operatorname{tr}\Psi = 1$ with the trace rounding nowhere, (SR1) and (SR2) — are finitely many exact identities on the emitted data (Theorem "Checkability and sufficiency of the source-replacement certificate" (transmitter/thm:certificate-sufficiency) (i)); (G3)(a) is decided by *reading the type of the datum each entry's right-hand-side slot holds, which is its tag*, and by nothing else — it compares no matrix against any other and constructs no right-hand side; (G3)(b) is a presence test. (G2)'s clause (SR3) and (G3)(b)'s record are attestations and are *not* decidable; they are tested for presence only;

(ii) *(outcome of a failure)* If any gate fails, the compiler emits no key-length-eligible declaration, and which of the *two* permitted outcomes it produces — refusal, or emission of a declaration that is not key-length-eligible — is fixed by the gate outcome table (Definition 9.3) and is not an implementer's choice; where the failures of one declaration meet the premises of a refusal row and of an emission row both, the outcome is the refusal. No protocol specification meets the premise of rows 2 and 4, the two rows of the gate outcome table that describe a certificate-absent case: by Definition 3.1 (a) a transmitter artefact in trusted-source mode at $d_A \geq 2$ carries a source-replacement certificate. The compiler never drops a constraint entry silently, never substitutes a default, never changes the source mode it was handed, and never synthesizes a licence.

(iii) *(no assertion)* Passing (G3)(b) records that an attestation was supplied. It asserts nothing about attack-set inclusion, and no claim of this contract does. Likewise passing (G2) records that the transmitter's licence record is present and that its five decided conditions hold of the emitted data: it asserts nothing about the apparatus, and what it buys for the run is bought under Assumption 10.3 — the certificate is decided once, for the one declaration that describes every round, which is where this module spends round homogeneity (Fact "Source replacement for prepare-and-measure protocols" (transmitter/fact:sr) drops the per-round superscript of the source state under exactly that assumption of the consumer's).

*The general format.* For a declaration in the general format of Definition 1.1 (DX1)–(DX6) the three parts hold with the following readings, of which the three above are the case of one Alice sector, one measured factor, POVM form, a pure-form certificate and no party-local entry.

(i′) *(decidability)* (G1) is decidable by Theorem 6.2 in its general-format reading. (G2) keeps its five conditions and is decidable in *one fur-*

*ther declared form* (Definition 9.1): the five conditions of a pure-form certificate, as above, a certificate declared in block form being refused at this gate; and, for an ensemble presentation, (E1), (E2), (E3) on the declared weighted ensemble together with the entry-for-entry identity of the declaration's Alice slots with the transposes of the declared blocks and of their declared-order sum — finitely many exact identities, comparisons of labels and integers, and finitely many summation decisions, the number fixed by $|\mathbb{T}|$, $|\mathcal{X}|$ and the declared dimensions. *Purity is decided by nothing there*: a presentation declares no $\Psi$, and purity of its blocks, (SR1) and (SR2) are theorems of the presentation (Lemma "Certificate defined by an ensemble presentation" (transmitter/lem:ensemble-presentation) (i), (ii)), the second giving every non-zero block as a positive multiple of a rank-one projector. *At one Alice sector it is not thereby waived*, the defined $\Psi$ being that one block, of unit trace, hence a rank-one projector of which the five conditions hold. *At two or more it is waived*, and the gate rests there on two statements — rank-one blocks by theorem, which is the hypothesis under which Theorem "Checkability and sufficiency of the source-replacement certificate" (transmitter/thm:certificate-sufficiency) (ii) holds in both directions, and that theorem's (iii), which reads purity nowhere — the first of which a general block form does not supply. (G3)(a) is decided by reading the type of the datum each entry's right-hand-side slot holds and, for the third form, the tag field it arrives with — three forms being in the licensed class, and still *nothing computed and nothing compared numerically.* (G3)(b) is a presence test, and (SR3) is tested for presence in every form.

(ii′) *(no row is added)* The outcome of every failure of the general format is fixed by a *row the gate outcome table already has.* A general-format check of Definition 6.1 that fails is (G1) and is row 1, the refusal naming the check. A certificate that is *present* and fails a condition of (G2) is row 3: a certificate that does not verify is a defect in the supplied data, not a weaker claim. *Row 3's premise is read of the declared form*, which reads the cell and adds no row: for a pure-form certificate the five conditions the cell enumerates are the conditions, and the cell is literal; for a certificate declared in block form the condition it fails is the form itself, (G2) admitting no block-form certificate (Definition 9.1), and the cell's premise — a certificate present and failing a condition of (G2) — covers it, whether or not it also fails $\Psi^2 = \Psi$; for an ensemble presentation the conditions are (E1) to (E3) and the identity of the declaration's Alice slots with the transposed declared blocks, and the cell's premise — a certificate present and failing a condition of (G2) — covers them, the enumeration in it being the pure form's list and not a restriction of the premise. *An ensemble*

*presentation accompanying a POVM-form declaration is a failure of that kind and is refused* (row 3), because nothing ties the declared $\{M_x\}$ to the presentation (Definition 9.1). A list carrying an entry declared general in trusted-source mode is row 5 and is emitted, not key-length-eligible; *a list carrying a party-local entry is not that row*, that form being licensed in either mode.

(iii′) *(no assertion)* Item (iii) holds as stated, and passing (G3)(b) records, rather than asserts, one further item: for a list carrying a party-local entry, what the attack-set licence record attests is a certificate of the module that produces the chain, on the scoped factor's own channel (Definition 2.5), and *no function of the emitted matrices decides it*, exactly as for the untagged forms. Passing (G2) on an ensemble presentation likewise records that the transmitter's record is present and that the conditions of *that form* hold of the emitted data, and asserts nothing about the apparatus; what it buys for the run is bought under Assumption 10.3 as in (iii), the certificate being decided once for the one declaration that describes every round.

**Proposition 9.6** (Dropping a trusted marginal enlarges the feasible set)**.** Let a specification request trusted-source mode with marginal $\tau_A$ satisfying D(ii), and a licensed list $\{(\hat{O}_k, \bowtie_k, c_k)\}_{k=1}^{m}$, $m \geq 0$, satisfying D(iii) — every $\hat{O}_k$ a block list against $\mathfrak{s}$ and Hermitian block by block, which the proof spends through Lemma 2.2 — with every entry declared homogeneous, so that the trusted-mode right-hand side it denotes is the constructed $c_k\tau_A$ (Definition 2.5). Let $\mathcal{W}^{\mathrm{tr}}$ be the feasible set of the trusted-mode declaration and $\mathcal{W}^{\mathrm{un}}$ that of the untrusted-mode declaration obtained by replacing the marginal slot with none and reading each entry's right-hand side as the scalar $c_k$, all other slots unchanged. Then

$$\mathcal{W}^{\mathrm{tr}} \subseteq \mathcal{W}^{\mathrm{un}}, \tag{9.1}$$

and consequently, for *any* functional $J$ on $\mathcal{L}(AQ)$, $\inf_{\mathcal{W}^{\mathrm{un}}} J \leq \inf_{\mathcal{W}^{\mathrm{tr}}} J$. The inclusion is *uniform in* $\tau_A$: it holds for every $\tau_A$ satisfying D(ii), and the untrusted set does not depend on which one the request carried.

## 10 Assumptions

**Assumption 10.1** [**A-SPEC**] (Specification faithful to the apparatus)**.** For a protocol specification $\mathfrak{S}$ (Definition 3.1): (i) the declared dimensions, alphabets and tables of $\mathfrak{S}$ describe the apparatus, and so do its declared measurements, namely: the physical receiver family $\{F_w^{\mathrm{ph}}\}_{w \in \mathcal{V}^{\mathrm{ph}}}$ (Definition 3.1 (b); in the general format, each measured factor's family of (b′)) is the measurement performed, in every round, on that round's physical receiver system alone, whatever happens in the other rounds, their outcomes

and public messages included; and where no source-replacement certificate is demanded of $\mathfrak{S}$ (in untrusted-source mode, or at $d_A = 1$), $\{M_x\}$ (Definition 3.1 (a)) is the measurement Alice performs on $A$; (ii) the key rule $r^0_{\mathsf{h}}$ is valued in $\mathcal{Z}$.

**Assumption 10.2** [**A-ALICE-MEAS**] (Alice's declared measurement where no source certificate is demanded)**.** (i) Where no source-replacement certificate is demanded of the protocol specification (in untrusted-source mode, or at $d_A = 1$), the letter $x \in \mathcal{X}$ Alice obtains in every round is the outcome of the declared measurement $\{M_x\}$ (Definition 3.1 (a)) applied to that round's system $A$ alone, the measurement she performs on $A$ being $\{M_x\}$ whatever happens in the other rounds, their outcomes and public messages included.

**Assumption 10.3** [**A-RHOM**] (Round homogeneity of the declaration)**.** (i) Every round of the protocol is described by the same specification, hence by the same declaration, the same source mode and the same operators: the family of the specification, the declaration, the source mode and the operators satisfies clause (i), sameness, of round homogeneity (Definition "Round homogeneity of a per-round family" (foundations/def:round-homogeneity)).

**Assumption 10.4** [**A-COIN**] (Public choices as private coins of declared bias)**.** (i) The declared public choices $t \in \mathcal{T}_{\text{pub}}$ are drawn from private random sources with their declared weights $q_t$, mutually independent, whichever party draws each, and, taken together, independent of the quantum systems and of the adversary; (ii) a public choice $f$ declared with a setting-dependent weighting $P(f \,|\, t)$ — the setting-dependent form of Definition 4.1, which may be declared at any measured factor — is drawn in each round by a party whose readout records the setting component $t$ of the outcome, from a private random source with the declared conditional distribution $P(\,\cdot \,|\, t)$ given that round's setting $t$; given $t$, $f$ is independent of the quantum systems, of the letter and every outcome of the round (the device outcome $y$ included), of every other declared public choice and of the adversary, taken together; (iii) every such choice is announced, or withheld, exactly as the public structure declares.

**Assumption 10.6** [**A-LINK**] (Soundness of every reduction link)**.** Each link of either segment of the reduction chain is sound: (i) for a *receiver* link, (a) where the link supplies no channel as data, only a theorem that constructs one, its channel exists as claimed; (b) it satisfies the pullback identity (7.1); and (c) security of the protocol using the emitted measurement implies security of the protocol using the physical one; (ii) for a *source* link, (a) the channel of the source identity (S3) of Definition "Source segment and its interface conditions" (transmitter/def:source-segment) exists as

claimed; and (b) security of the protocol preparing the virtual ensemble implies security of the protocol preparing the specified one; (iii) for a link of either segment, every reachable state after the reduction satisfies every constraint entry the link transports; (iv) a link of either segment that supplies its channel *as data* supplies a completely positive and trace-preserving map.

**Assumption 10.7** [**A-ATK**] (Attack-set licence)**.** For a non-empty constraint list $\mathfrak{L}$ that is licensed for the declared source mode (Definition 2.5) and backs a key-length claim: (i) the reduction chain that produced the declaration certifies the *licensed reduction*: security of the physical protocol reduces to security of the consuming engine's generated round model (Definition "Round model and licensed attack sets" (meat-core/def:roundmodel)) with attack sequences restricted to the $\mathfrak{L}$-admissible ones, that is, every attack available against the physical protocol whose registers are finite-dimensional and whose first channel receives every transmitted system is represented, after the reduction, by a sequence whose per-round channel $A_j$ lies, for every round $j$, in the restricted set $\mathfrak{A}_{\mathfrak{L}}(E_{j-1}, QE'_j)$ of that definition, with no $A$ output, in trusted-source mode, and in its untrusted-source-mode counterpart, with $A$ supplied by the adversary, in untrusted-source mode.

**Assumption 10.8** [**A-IDEAL**] (Ideal qubit receiver on the Band-A route)**.** In a Band A specification: (i) the declared receiver is exactly an ideal projective qubit measurement; (ii) it has unit detection efficiency, no dark counts, no loss, no multi-photon events and no misalignment; (iii) the transmitted system is exactly a qubit.

# Engine requirements

## 11 Requirements: the emitted declaration

Requirement 11.1 (Machine keys of the emitted declaration). The engine emits the sector map $\mathfrak{s}$ of a protocol declaration as three machine keys, `sector_index_set` for its ordered block identifiers, `sector_dims` for their dimensions and `sector_block_kind` for its block kind, so that a consumer comparing sector maps can compare all three and not two of them. It emits the label-to-index map $\mathsf{idx}$ of Definition 3.3 alongside the declaration, under `label_index_map`, and with each constraint entry the tag of its right-hand side, under `constraint_rhs_tag`, so that the consuming engine's type check has a datum (Definition 2.5). The sifting predicate $\varsigma$ and the set $\widehat{\mathcal{C}}_{\mathrm{key}}$ of key-live values are not emitted.

Requirement 11.2 (Operators on the measured system held as block lists). The engine holds and emits every operator on $Q$ as its block list against the sector map $\mathfrak{s}$ (Definition 1.1). Where $|\mathbb{S}| > 1$ it does not materialize the direct sum $\bigoplus_s F^{(s)}$ of such a list, with one exception: condition (C4) of Definition 7.1 forms it as the argument of that one decision, decides it and discards it. The convention `operator_form` states this rule and its scope.

Requirement 11.3 (Declared data passed through unchanged). The engine passes through unchanged the declared data that the compilation map reads into a declaration: the transmitter artefact's marginal slot, the value [1] the transmitter filled it with at $d_A = 1$ included; Alice's measurement slot, in the form the artefact declares, which it changes only by weighting it in that form (Definition 4.1) where a choice family is presented; the transcript; and each constraint entry's right-hand-side datum and, where the entry carries one, its scope. It fills no slot and demands no certificate of its own. It constructs no right-hand side of an emitted constraint entry: it forms no $c_k\tau_A$ for one and reads no $c_k$ off any matrix. It converts neither form of Alice's measurement slot into the other: it forms neither (1.7) from an ensemble slot nor $\tau_A^{1/2} M_x \tau_A^{1/2}$ from a POVM slot.

Requirement 11.4 (Per-factor decisions on the declared blocks). For a declaration in the general format of Definition 1.1: no entry of a product family (1.5) is formed as a datum, rounded or decided on, here or downstream; the lifted operator of a scoped entry is never formed as a datum; and the family (1.7) that an ensemble form denotes is formed by no one and decided by nothing. Every exact predicate is decided per measured factor, on that factor's blocks and in that factor's declared order, and on Alice's data per Alice sector (Definition 1.3). The engine refuses by type a presentation of a choice-weighted family indexed by a product alphabet (Definition 4.1).

Requirement 11.5 (Reading the tag of a constraint entry). The engine reads the tag of an entry of one of the two untagged forms of Definition 2.5 off the type of its right-hand-side datum. Where such an entry arrives carrying a tag field, check V6 of Definition 6.1 compares that field with the type of the datum it names and refuses a disagreement; the tag is never taken on the upstream module's word against the object the entry holds. An entry is read as party-local exactly when it arrives carrying that tag; check V6 then requires its right-hand-side slot to hold a real scalar and its scope to name a measured factor of the emitted sector map, and refuses otherwise. No tag is inferred: an entry arriving without the party-local tag is never read party-locally, whatever module emitted it.

Requirement 11.6 (Exact decisions without a tolerance). The engine declares no numerical tolerance, and there is no window in this contract at which anything could be decided: no check of the validator (Definition 6.1) reads a window or is decided at one, and licensing (Definition 2.5) and the predicates of Definition 1.3 are decided exactly.

Requirement 11.7 (The source-replacement licence passed on with the declaration). The engine passes the transmitter's source-replacement licence record `source_replacement_license` (Definition "Source-replacement certificate" (transmitter/def:srcert)) on to the consuming engine with a declaration exactly when it emits that declaration in trusted-source mode at $d_A \geq 2$, after gate (G2) of Definition 9.1 has re-decided its conditions on the emitted data; the certificate and the attestation (SR3) it carries are the ones supplied, and the engine asserts no attestation of its own (Requirement "Pass-through of the retention attestation" (transmitter/req:attestation-record)). At $d_A = 1$ the record is neither demanded nor emitted (Requirement "Certificate and attestation demanded only at $d_A \geq 2$" (transmitter/req:licence-scope)).

Requirement 11.8 (Letter-independent emission at a one-dimensional trusted A). For a specification in trusted-source mode at $d_A = 1$, the engine decides, for every $x \in \mathcal{X}$ in the declared order and exactly on the declared numbers, the identity

$$\sigma_x \;=\; m_x \sum_{y \in \mathcal{X}} \sigma_y,$$

where $m_x$ is the one entry of Alice's measurement element $M_x$, read as the declaration carries it (Definition 3.3: for a choice-weighted family the weighted element the engine forms; in the ensemble form, $N_x$, which equals $M_x$ at $d_A = 1$ by (1.7)), and $\sigma_x = p_x \rho_x^{\mathrm{v}}$ is the weighted member for $x$ of the virtual ensemble the transmitter artefact emits, which at an empty source segment is the specified one (Definition "Source segment and its interface conditions" (transmitter/def:source-segment)); where the artefact emits the ensemble by its weighted members, $\sigma_x$ is the declared member. Products and sums are formed in exact arithmetic, entry by entry. A specification for which the identity fails at some letter is refused, before a declaration is emitted, with `da1_source_identity_failed`, naming the first such letter in the declared order and the first entry at which the two sides differ.

Requirement 11.9 (Emission of the weighted families and the weighting record). The engine emits both forms of the choice-weighted family of Definition 4.1, each where it is the emitted family: the setting-independent form (4.1) on either party's side, and the setting-dependent form (4.3) at a measured system, of whichever party, after that system's receiver segment.

A setting-independent presentation of a measuring party's physical receiver family is the emitted family only where that party's receiver segment delivers it as its target POVM; in front of any other segment it is handed to the chain as the physical receiver family of Definition 3.1 (b), and the emitted family is the chain's. For the setting-dependent relabelling it emits no link and no register for the choice value, and the chain record it consumes and re-emits is unchanged by the weighting; the weightings are recorded beside the chain in the emitted artefact, in `choice_weighting_record`, one entry per measured factor at which a weighting is applied, each entry naming that factor; like the setting-independent choice-weighted family, they are this module's own construction.

## 12 Requirements: choice weighting

Requirement 12.1 (Decision on each declared conditional family). Wherever a choice-weighted presentation is read, on either party's side, the engine decides, of each choice value $t \in \mathcal{T}_{\mathrm{pub}}$ separately and on the declared numbers, the condition Definition 4.1 places on the conditional family $\{N_{u|t}\}_u$ — in POVM form, its exact admissibility in the reading of that definition; in ensemble form, the condition that definition states in its place: (H) and (P) per block, the ensemble identity in place of (C), and (T) — before it forms any product and before any condition of Definition 7.1 is decided on the chain: wherever the presented physical family is the weighted one, that decision and the representability decision on the products it permits precede the consumption of the chain, so that no condition on the chain is decided on a product that was not decided first. A failure is refused with `conditional_family_not_admissible`, naming the declaring party, and the choice value, the predicate, the block and the index of the *first* failure — for (S), the index of the first inexact addition; in ensemble form the predicate named is the ensemble identity where that is the one that failed. The decision stops at the first failure and does not enumerate the rest.

Requirement 12.2 (Declared weights as binary64 input). The weights of either form of Definition 4.1 — each $q_t$ of the setting-independent form and each entry of $P$ of the setting-dependent form — are binary64 numbers. The engine rejects as malformed input a declared weight, in either form, that is not exactly representable as a finite binary64 number, before that weight is used in an accumulation or a product. This is the declared input type: the rejection is not a failed summation predicate, and it adds no row of the gate report and no refusal string.

Requirement 12.3 (Preparation of user weights before the module boundary). Ordinary decimal and rational user inputs for the weights of Definition 4.1 may be converted and prepared by the declaring entry point before

they cross the module boundary into this module. An entry point that does so records the supplied and the prepared values and their displacement, preserves normalisation and every required bound direction, and leaves this module's exact tests unchanged. The preparation asserts no bound on a resulting change of the key rate.

Requirement 12.4 (Setting-dependent weighting read through the setting only). The engine refuses by type, with `choice_weight_depends_on_outcome`, a weighting of the setting-dependent form of Definition 4.1 that is presented indexed by the outcome alphabet $\mathcal{V}$, by the device outcome, or by any component of the outcome other than its setting; in the general format, one presented indexed by a measured factor's whole outcome, by another factor's outcome, or by any component other than that factor's setting.

Requirement 12.5 (Decision on the declared weight list). The engine decides the conditions of Definition 4.1 on the declared weights $(q_t)_{t\in\mathcal{T}_{\mathrm{pub}}}$ of a choice-weighted presentation: each $q_t > 0$ by its sign, and $\sum_t q_t = 1$ exactly, with the sum accumulated in the declared order of $\mathcal{T}_{\mathrm{pub}}$ — the order in which the declared public-choice record lists it (Requirement "Summation order declared with the data" (foundations/req:declared-order)) — and required to round nowhere. A declared weight that is zero or negative is refused by type. A weight list whose exact sum is not 1, or whose declared-order accumulation rounds at some addition, is refused with `choice_weights_not_normalized`, and no declaration is emitted.

Requirement 12.6 (Stochastic matrix of a setting-dependent weighting). The engine decides the conditions of Definition 4.1 on the declared matrix $P$ of the setting-dependent form: every entry $P(f\,|\,t) \geq 0$ by its sign, and every row sum $\sum_f P(f\,|\,t) = 1$ exactly, accumulated in the declared order of the choice alphabet (Requirement "Summation order declared with the data" (foundations/req:declared-order)) and required to round nowhere; the support of the weighting is decided by the sign of each entry. A negative entry, or a row whose declared-order sum is not exactly 1, is refused with `choice_weight_not_stochastic`.

Requirement 12.7 (Emission decisions on the weighted products). The engine forms the products $q_t N_{u|t}$ and $P(f\,|\,t(v))F_v$ of Definition 4.1 exactly, on the declared dyadic numbers, and takes the two emission decisions of that definition before emitting anything. The first, that every entry of every product it forms is a binary64 number, is taken on every product the map forms, whether the product is emitted or handed on: the intermediate families of an iterated presentation are products, and a presentation on Bob's side before the chain produces the physical family that (C2) and (C4) of

Definition 7.1 read and that, in Band A, (C3) decides, so no condition of the chain is decided on a product whose entries were not decided first. The second is that the emitted family satisfies Definition 1.3. If either decision fails, the engine refuses — with `emitted_value_not_representable` in the first case and with the predicate that failed in the second — and emits no declaration. `emitted_value_not_representable` covers exactly an exact product of declared numbers that is not a binary64 number: an entry that is binary64 but lies off whatever lattice the instance declared its own data on is emitted, and the five conditions the consumer states are decided on it exactly and on it alone.

Requirement 12.8 (No lattice exponent and no snapping of the emission). The engine declares no lattice exponent for the families it emits. It establishes the five conditions of Definition 1.3 of those families directly, on the numbers it is about to emit, and refuses where one fails (Requirement 12.7); it neither rounds nor snaps its own emission, and it never declares a nearby family (`no_snapping`). A specification whose weighting does not land exactly on the lattice is a specification to be re-declared at the instance. What the engine owes its consumer is the five conditions, and it decides them rather than inheriting them from a lattice.

Requirement 12.9 (No fixed fraction outside the measurement operators). The emitted declaration carries the output-state test-statistics model (Assumption "Output-state test statistics" (reductions/ass:tsm)) and never the fixed-fraction model: no path of the engine produces an artefact carrying a fixed-fraction model, and no weight $q_t$ is emitted outside a POVM element. A specification requesting the fixed-fraction encoding of Theorem 4.4 (iv) — the conditional families together with a separate state-independent fraction $q_t$ to be applied outside the operators — is refused with `fixed_fraction_requested`.

## 13 Requirements: validation, feasibility and the chain

Requirement 13.1 (Refusals of the validator's checks). The engine runs checks V1 to V8 of Definition 6.1 on every candidate declaration and refuses a candidate that fails one, emitting no declaration, with a reason naming the failing check. It repairs nothing and completes no declared table with a default value. Where a check has a reason of its own, the refusal carries that reason and what the check names with it:

(i) V1: `sector_map_mismatch`, naming the first position at which the compared lists differ, and in the general format, for the block kind across the factors, the first factor at which the kind differs;

(ii) V2: `alice_marginal_trace_rounds` where an addition of the marginal's trace rounds; in the general format `alice_sector_marginal_zero`, naming the sector, where a block of the marginal is zero beside a measurement slot in ensemble form;

(iii) V3: `alice_povm_summation_rounds`, in the general format carrying the Alice sector; `bob_povm_summation_rounds`, carrying the sector and the outcome index at which the first inexact addition occurred, and in the general format the factor as well; in the general format `alice_ensemble_marginal_mismatch`, carrying the sector and the first differing entry, `alice_ensemble_sum_rounds`, carrying the sector and the index of the first inexact addition, and `ensemble_form_in_untrusted_mode` for a slot in ensemble form beside a marginal slot holding `none`;

(iv) V6: in the general format `constraint_scope_not_a_factor`, naming the entry;

(v) V7: `no_feasibility_witness`, naming, where a supplied witness was rejected on its map, the first position at which the two lists differ.

Requirement 13.2 (Transcript of a check that stops early). A check of Definition 6.1 may stop after a failed prerequisite or predicate. Its transcript reports only the decisions the check actually took, in their execution order, and the engine infers no outcome for a predicate it did not execute.

Requirement 13.3 (Default candidate before a supplied witness). Where the constraint list is non-empty, on a candidate declaration satisfying D(i)–D(iii) and D(vi), which checks V1–V3 and V6 of Definition 6.1 decide (a candidate failing one of them is refused there), the engine decides the conditions of (2.1), in the general format those of (2.3), first on the default candidate (5.2), in the general format (5.3), and, if it fails, on a supplied witness $\omega^\star$, read from the input `feasibility_witness`. It forms the default candidate only to test it, and it emits neither candidate. It refuses with `no_feasibility_witness` exactly when the default candidate fails and either no witness is supplied or the supplied one fails.

Requirement 13.4 (Reporting a refusal for want of a feasibility witness). The engine never reports a refusal with `no_feasibility_witness` to a user as a defect in the protocol.

Requirement 13.5 (Decision on the presented physical receiver family). On every route the engine decides, on the declared numbers, the condition clause (iv) of Definition 3.2 places on the physical receiver family as presented, in the general format on each measured factor's: presented directly,

(S), (H), (P) and (C) of Definition 1.3 on its physical receiver space as one block, in the declared order of its alphabet; presented as a choice-weighted family, condition (S) on its own declared-order sum, after the decision of Requirement 12.1 on its conditional families and the representability decision on the products formed from them. It decides it before any condition of Definition 7.1 is decided on the chain, and refuses a failure with `physical_family_not_admissible`, naming the measured factor, the predicate and the index of the first failure (for (S), the index of the first inexact addition). The decision stops at the first failure.

Requirement 13.6 (Refusals of the interface conditions on a chain). The engine decides the interface conditions on a chain of Definition 7.1 where the chain is consumed, before a declaration exists, and refuses a chain that fails one; no refusal emits a declaration.

(i) A chain failing one of (C1)–(C5) or (S1)–(S4) is refused with `chain_interface_condition_failed`, naming the condition and, for (C1)–(C5) in the general format, the factor, and, for (C4) and (S3), the outcome or signal index at which the first entry disagreed. A non-empty source segment failing one of (A1)–(A3) is refused.

(ii) Beside a non-empty source segment, an entry declared general is refused with `general_entry_beside_source_segment`, naming the entry.

(iii) A declaration carrying a non-empty source segment in untrusted-source mode is refused with `source_segment_in_untrusted_mode`.

Requirement 13.7 (Band A only within the ideal qubit receiver). The engine recognises the Band A route on the record, reading the shape that condition (b) of Definition 8.1 gives the chain and the sector clause of its (a), for a specification whose source segment is empty, whose emitted sector set (in the general format, the joint one) is a singleton, and in which, for some measured factor, the receiver segment is one link with status exact, equal source and target descriptors, a channel that, where the link supplies one as data, is exactly the identity map on its system, an empty constraint slot and an empty licence slot; it decides the link condition on the chain record of every measured factor. The clauses of (b) this recognition leaves open — that the link supplies its channel as data, its outcome correspondence and its target POVM — are decided by Requirement 13.8. It refuses such a specification, before a declaration is emitted, with `band_a_outside_ideal_receiver`, naming the first clause below that fails and, for (iii), the choice value where there is one and the outcome index of the first element that is not a projector, unless

(i) the specification declares exactly one measured factor;

(ii) that factor's physical receiver system has dimension two and, in trusted-source mode, the transmitted system of the transmitter artefact (Definition 3.1 (a)) has dimension two; and

(iii) that factor's physical receiver family as presented is projective, decided exactly on the declared numbers: presented directly, every element satisfies $F_w^{\mathrm{ph}} F_w^{\mathrm{ph}} = F_w^{\mathrm{ph}}$; presented as a choice-weighted family, every element of every declared conditional family satisfies $N_{u|t} N_{u|t} = N_{u|t}$ (Definition 4.1).

Requirement 13.8 (Band A only on an identity link that delivers the declared family). For a specification that Requirement 13.7 recognises on the Band A route and does not refuse, the engine decides on the chain record of its one measured factor, before a declaration is emitted, that

(i) the one link of that factor's receiver segment supplies its channel as data, which that recognition has then decided to be exactly the identity map on its system; and

(ii) the outcome correspondence of that segment is the identity: the physical receiver alphabet $\mathcal{V}^{\mathrm{ph}}$ and the alphabet $\mathcal{V}$ of the link's target POVM are one set, and $\gamma(w) = w$ for every $w \in \mathcal{V}^{\mathrm{ph}}$;

and refuses a specification failing either, before a declaration is emitted, with `band_a_link_not_identity`, naming the first clause that fails. On a specification passing both, condition (C4) of Definition 7.1 is decided, as on every chain whose links supply their channel as data, and reads $F_v = F_v^{\mathrm{ph}}$ for every $v \in \mathcal{V}$, entry for entry, $\{F_w^{\mathrm{ph}}\}$ being the physical receiver family as presented and, for a choice-weighted family, the family of products Definition 4.1 forms; a failure of it is refused as every failure of (C4) is (Requirement 13.6). A declaration emitted from a specification so recognised therefore satisfies Definition 8.1 (b) on its record — its one link the identity, supplied as data, its outcome correspondence the identity and its target POVM the declared family — and the rules that read the route, the ledger's among them (Requirement 13.10), read it of such a declaration alone.

Requirement 13.9 (Emission of both chain segments and every link slot). The engine emits the chain record (`reduction_chain_record`) with both of its segments, the source segment and the receiver segment, and on every link with both its constraint slot and its licence slot, each whether or not it is empty. This includes Band A, where the source segment is empty and the receiver segment is one identity link with both slots empty.

Requirement 13.10 (Omission of the link-soundness row only on a proved chain). The engine omits the row of Assumption 10.6 from the assumption ledger of a compilation only (i) on the Band A route, for a declaration whose record satisfies Definition 8.1 (b) as Requirements 13.7 and 13.8 decide it — the one link the identity, supplied as data, its outcome correspondence the identity and its target POVM, by the decided (C4), the presented physical family — where Theorem 8.2 (v) proves every clause of it; or (ii) where the source segment is empty and, for every measured factor of the specification, all of the following hold: its receiver segment consists of one link, the instance link of an artefact that the reductions module emitted in the same run, with status *exact* and no channel supplied as data; the outcome correspondence of the segment is the identity on that factor's target alphabet; and that factor's physical receiver family as presented is exactly the compression $\{\iota^\dagger M'^{\mathrm{ph}}_v \iota\}_v$ of that artefact's declared model onto the presented physical receiver space, $\iota$ its isometric inclusion into the physical system, or the declared model itself. At several measured factors the omission rests in addition on A-MEAS (Assumption "Each measuring party's device as one POVM per round" (reductions/ass:meas)) (iv), that the round's physical measurement is the product of the factors' measurements. On either route, at a factor whose delivered family is weighted after its segment, in the setting-dependent form of Definition 4.1, the omission rests in addition on Lemma "Linearity: post-processing, private settings and mixtures" (reductions/lem:linear) (i): the weighting of the family the segment's link delivers and the identically weighted physical family stand in the squashing relation under the same channel, so the transfer applies to the weighted protocol as to the unweighted one; on route (i) the channel is the identity and the two weighted families are the same matrices (Theorem 8.2 (v)). That each link is that artefact's instance link and that each presented family is that compression is established by the part of the engine that ran the reductions module and docked its artefacts, only for an instance for which Remark 7.7 names the claim showing the docked family to be that compression, and is passed to the compiler beside the specification (`chain_provenance`); the compiler never reads it from the specification. The compiler decides, on the presented chain, the parts of (ii) that are data: for every measured factor the length of its segment, the status tag, the channel slot and the outcome correspondence (Definition 7.1, (C2)), and the length of the source segment. Where condition (ii) is not passed, or a part of it decided on the chain fails, the row is emitted.

Requirement 13.11 (The reference companion compiled beside the declaration). Where the last link of the receiver segment of one or more measured factors was emitted with a reference companion (Requirement "Emitting the reference companion beside the artefact" (reductions/req:reference-companion)),

the engine emits, beside the declaration $\mathfrak{D}$ it compiles, one *compiled reference companion* $\mathfrak{D}^{\mathrm{ref}}$, in `reference_companion`:

(i) it is the tuple the compilation map (Definition 3.3) gives from the same specification with, for every measured factor whose last link was so emitted, that link's target family replaced by its companion's and that link's entries replaced by its companion's relaxed entries (Lemma "A rounded constraint entry with its constant relaxed" (reductions/lem:row-relaxation)), those of the entries requested of the link included where the companion carries them, and none in place of a requested entry the companion does not carry (Requirement "Emitting the reference companion beside the artefact" (reductions/req:reference-companion) (iii)); the engine forms it by the code path that forms $\mathfrak{D}$, every other slot read from the same data, the family and the entries of every other measured factor's last link among them;

(ii) where a measured factor's family is weighted after its chain, by that factor's weighting (Definition 4.1), every product $P(f \mid t(v))\, F_v$ is formed exactly from the companion's enclosure: the centre multiplied by the weight and every radius by its modulus, so that the result encloses the product of the weight and the companion's element. Where the companion carries a factor enclosure (Definition "Factor enclosure of a block" (foundations/def:factor-enclosure)) of a block of $F_v$ (Requirement "Emitting the reference companion beside the artefact" (reductions/req:reference-companion) (vii)), the block of the product carries, beside its entrywise enclosure, the one of Lemma 4.3: the same box, its centres and radii unaltered, and the core multiplied by the weight in exact rational arithmetic. Where the family is not weighted after its chain, every factor enclosure the companion carries is passed to $\mathfrak{D}^{\mathrm{ref}}$ unaltered;

(iii) the engine decides on $\mathfrak{D}^{\mathrm{ref}}$ none of the predicates it decides on the binary64 numbers of $\mathfrak{D}$: no Hermiticity, positivity or completeness of the family, no completeness sum in a declared order (Definition 1.3), no chain condition and no check of the validator; it passes the companion's entries through as Requirement 11.3 passes every entry;

(iv) it emits no bundle of a run of its own from $\mathfrak{D}^{\mathrm{ref}}$ and marks it evaluation-only, a declaration with no run of its own, with a reference to the declaration $\mathfrak{D}$ it is the companion of and the companion's record carried unchanged.

Where no measured factor's last link carries a companion, nothing of this requirement applies, and the run forms its first-order number on $\mathfrak{D}$ alone, its blocks resting as the records of the artefact say.

## 14 Requirements: the emission gates

Requirement 14.1 (The key-length eligibility flag). The engine emits the flag `key_length_eligible` true for a declaration only if that declaration is key-length-eligible (Definition 9.1).

Requirement 14.2 (Outcome of each gate failure). For each gate failure the engine produces the outcome the gate outcome table (Definition 9.3) fixes for it, and no other; where the failures of one declaration meet the premises of a refusal row and of an emission row both, it produces the refusal:

(i) row 1, a check that fails: a refusal naming the check, which in the general format covers `ensemble_form_in_untrusted_mode`, `alice_ensemble_marginal_mismatch`, `alice_ensemble_sum_rounds`, `alice_sector_marginal_zero` and `constraint_scope_not_a_factor`;

(ii) row 3, trusted-source mode at any $d_A$ with a certificate present and failing a condition of gate (G2) for its form: a refusal naming the condition, an ensemble presentation beside a POVM-form declaration being refused with `ensemble_presentation_without_ensemble_form`;

(iii) rows 5 and 6, a constraint list not in the licensed class or the attack-set licence record `attack_set_license` absent: an emission with `key_length_eligible` false.

*A trusted-mode specification at $d_A \geq 2$ whose certificate is absent does not reach this module.* The mode is read off the transmitter artefact's marginal slot (Definition 3.3), and the transmitter refuses a trusted-mode request at $d_A \geq 2$ that carries no certificate before any artefact is emitted, with `trusted_mode_without_certificate` (Requirement "Certificate and attestation demanded only at $d_A \geq 2$" (transmitter/req:licence-scope); the modes are defined in Definition "Source specification" (transmitter/def:source-spec)). So an artefact in trusted mode at $d_A \geq 2$ carries the record gate (G2) tests for, and the two rows of the table that describe a certificate-absent case — rows 2 and 4, at $m = 0$ and at $m \geq 1$ — are *unreachable through the transmitter artefact*. At $d_A = 1$ the gate demands no certificate at all, the slot holds the [1] the transmitter filled it with (Requirement "The unique marginal served at $d_A = 1$" (transmitter/req:marginal-fill)), and one supplied there and failing is refused by row 3, its scope being every $d_A$. Rows 2 and 4 are unreachable in the general format for the same reason, which reads the transmitter's refusal and no property of the general format. The engine never supplies a missing attestation and never proceeds as though it had one.

Requirement 14.3 (Attestations tested for presence only). The engine tests the attestations of Definition 9.1 for presence only, never for truth: at gate (G2) the clause (SR3) of the source-replacement licence record `source_replacement_license` (Definition "Source-replacement certificate" (transmitter/def:srcert)), in every form of the certificate, and at gate (G3)(b) the attack-set licence record `attack_set_license`, the attestation of Assumption 10.7. It never supplies an attestation and never verifies one.

Requirement 14.4 (A vacuous pass recorded as vacuous). Where gate (G3)(a) of Definition 9.1 holds vacuously, in untrusted-source mode at any $d_A$, the engine records it in the gate report `gate_report` as *passed vacuously* rather than *passed*, and it never reports the vacuous pass as evidence of anything.

## References

[1] Devashish Tupkary, Shlok Nahar, Amir Arqand, Ernest Y.-Z. Tan, and Norbert Lütkenhaus. A rigorous and complete security proof of decoy-state BB84 quantum key distribution, 2026.